

XD1250, XD1600, XD2000, XD2500, XD3600, XD4150, XD5000

Instruction book

Atlas Copco

XD1250, XD1600, XD2000, XD2500, XD3600, XD4150, XD5000

Instruction book

Original instructions

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1 SAFETY PRECAUTIONS

1.1 SAFETY ICONS

EXPLANATION

	Danger for life
	Warning
	Important note

1.2 SAFETY PRECAUTIONS, GENERAL

GENERAL PRECAUTIONS

1. The operator must employ safe working practices and observe all related work safety requirements and regulations.
2. If any of the following statements does not comply with the applicable legislation, the stricter of the two shall apply.
3. Installation, operation, maintenance and repair work must only be performed by authorized, trained, specialized personnel.
4. The compressor is not considered capable of producing air of breathing quality. For air of breathing quality, the compressed air must be adequately purified according to the applicable legislation and standards.
5. Before any maintenance, repair work, adjustment or any other non-routine checks, stop the compressor, press the emergency stop button, switch off the voltage and depressurize the compressor. In addition, the power isolating switch must be opened and locked.
6. On units powered by a frequency converter, wait 10 minutes before starting any electrical repair.

	If the machine is equipped with an automatic restart after voltage failure function and if this function is active, be aware that the machine will restart automatically when the
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7. Never play with compressed air. Do not apply the air to your skin or direct an air stream at people. Never use the air to clean dirt from your clothes. When using the air to clean equipment, do so with extreme caution and wear eye protection.
8. The owner is responsible for maintaining the unit in safe operating condition. Parts and accessories shall be replaced if unsuitable for safe operation.

9. It is prohibited to walk or stand on the unit or on its components.

1.3 SAFETY PRECAUTIONS DURING INSTALLATION



All responsibility for any damage or injury resulting from neglecting these precautions, or non observance of the normal caution and care required for installation, operation, maintenance and repair, even if not expressly stated, will be disclaimed by the

PRECAUTIONS DURING INSTALLATION

1. The machine must only be lifted using suitable equipment in accordance with the applicable safety regulations. Loose or pivoting parts must be securely fastened before lifting. It is strictly forbidden to dwell or stay in the risk zone under a lifted load. Lifting acceleration and deceleration must be kept within safe limits. Wear a safety helmet when working in the area of overhead or lifting equipment.
2. The unit is designed for indoor use. If the unit is installed outdoors, special precautions must be taken; consult your supplier.
3. In case the device is a compressor, place the machine where the ambient air is as cool and clean as possible.

If necessary, install a suction duct. Never obstruct the air inlet. Care must be taken to minimize the entry of moisture at the inlet air.

4. Any blanking flanges, plugs, caps and desiccant bags must be removed before connecting the pipes.
5. Air hoses must be of correct size and suitable for the working pressure. Never use frayed, damaged or worn hoses. Distribution pipes and connections must be of the correct size and suitable for the working pressure.
6. In case the device is a compressor, the aspirated air must be free of flammable fumes, vapors and particles, e.g. paint solvents, that can lead to internal fire or explosion.
7. In case the device is a compressor, arrange the air intake so that loose clothing worn by people cannot be drawn in.
8. Ensure that the discharge pipe from the compressor to the aftercooler or air net is free to expand under heat and that it is not in contact with or close to flammable materials.
9. No external force may be exerted on the air outlet valve; the connected pipe must be free of strain.
10. If remote control is installed, the machine must bear a clear sign stating: DANGER: This machine is remotely controlled and may start without warning.

The operator has to make sure that the machine is stopped and depressurized and that the electrical isolating switch is open, locked and labelled with a temporary warning before any maintenance or repair. As a further safeguard, persons switching on or off remotely controlled machines shall take adequate precautions to ensure that there is no one checking or working on the machine. To this end, a suitable notice shall be affixed to the start equipment.

11. Air-cooled machines must be installed in such a way that an adequate flow of cooling air is available and that the exhausted air does not recirculate to the compressor air inlet or cooling air inlet.
12. The electrical connections must correspond to the applicable codes. The machines must be earthed and protected against short circuits by fuses in all phases. A lockable power isolating switch must be installed near the compressor.
13. On machines with automatic start/stop system or if the automatic restart function after voltage failure is activated, a sign stating "This machine may start without warning" must be affixed near the instrument panel.
14. In multiple compressor systems, manual valves must be installed to isolate each compressor. Non-return valves (check valves) must not be relied upon for isolating pressure systems.
15. Never remove or tamper with the safety devices, guards or insulation fitted on the machine. Every pressure vessel or auxiliary installed outside the machine to contain air above atmospheric pressure must be protected by a pressure relieving device or devices as required.
16. Piping or other parts with a temperature in excess of 70°C (158°F) and which may be accidentally touched by personnel in normal operation must be guarded or insulated. Other high temperature piping must be clearly marked.
17. For water-cooled machines, the cooling water system installed outside the machine has to be protected by a safety device with set pressure according to the maximum cooling water inlet pressure.
18. If the ground is not level or can be subject to variable inclination, consult the manufacturer.
19. If the device is a dryer and no free extinguishing system is present in the air net close to the dryer, safety valves must be installed in the vessels of the dryer.



Also consult following safety precautions: [Safety precautions during operation](#) and [Safety precautions during maintenance](#).

These precautions apply to machinery processing or consuming air or inert gas. Processing of any other gas requires additional safety precautions typical to the application which are not included herein.

1.4 SAFETY PRECAUTIONS DURING OPERATION



All responsibility for any damage or injury resulting from neglecting these precautions, or non observance of the normal caution and care required for installation, operation, maintenance and repair even if not expressly stated will be disclaimed by the

PRECAUTIONS DURING OPERATION

1. Never touch any piping or components of the compressor during operation.

2. Use only the correct type and size of hose end fittings and connections. When blowing through a hose or air line, ensure that the open end is held securely. A free end will whip and may cause injury. Make sure that a hose is fully depressurized before disconnecting it.
3. Persons switching on remotely controlled machines shall take adequate precautions to ensure that there is no one checking or working on the machine. To this end, a suitable notice shall be affixed to the remote start equipment.
4. Never operate the machine when there is a possibility of taking in flammable or toxic fumes, vapors or particles.
5. Never operate the machine below or in excess of its limit ratings.
6. Keep all bodywork doors shut during operation. The doors may be opened for short periods only, e.g. to carry out routine checks. Wear ear protectors when opening a door.

On compressors without bodywork, wear ear protection in the vicinity of the machine.

7. People staying in environments or rooms where the sound pressure level reaches or exceeds 80 dB (A) shall wear ear protectors.
8. Periodically check that:
 - All guards are in place and securely fastened
 - All hoses and/or pipes inside the machine are in good condition, secure and not rubbing
 - No leaks occur
 - All fasteners are tight
 - All electrical leads are secure and in good order
 - Safety valves and other pressure relief devices are not obstructed by dirt or paint
 - Air outlet valve and air net, i.e. pipes, couplings, manifolds, valves, hoses, etc. are in good repair, free of wear or abuse
 - Air cooling filters of the electrical cabinet are not clogged
9. If warm cooling air from compressors is used in air heating systems, e.g. to warm up a workroom, take precautions against air pollution and possible contamination of the breathing air.
10. On water-cooled compressors using open circuit cooling towers, protective measures must be taken to avoid the growth of harmful bacteria such as *Legionella pneumophila* bacteria.
11. Do not remove any of, or tamper with, the sound-damping material.
12. Never remove or tamper with the safety devices, guards or insulations fitted on the machine. Every pressure vessel or auxiliary installed outside the machine to contain air above atmospheric pressure shall be protected by a pressure relieving device or devices as required.
13. Yearly inspect the air receiver. Minimum wall thickness as specified in the instruction book must be respected. Local regulations remain applicable if they are more strict.



Also consult following safety precautions: [Safety precautions during installation](#) and [Safety precautions during maintenance](#).

These precautions apply to machinery processing or consuming air or inert gas. Processing of any other gas requires additional safety precautions typical to the application which are not included herein.

1.5 SAFETY PRECAUTIONS DURING MAINTENANCE OR REPAIR



All responsibility for any damage or injury resulting from neglecting these precautions, or non-observance of the normal caution and care required for installation, operation, maintenance and repair, even if not expressly stated, will be disclaimed by the

PRECAUTIONS DURING MAINTENANCE OR REPAIR

1. Always use the correct safety equipment (such as safety glasses, gloves, safety shoes, etc.).
2. Use only the correct tools for maintenance and repair work.
3. Use only genuine spare parts.
4. All maintenance work shall only be undertaken when the machine has cooled down.
5. A warning sign bearing a legend such as "Work in progress; do not start" shall be attached to the starting equipment.
6. Persons switching on remotely controlled machines shall take adequate precautions to ensure that there is no one checking or working on the machine. To this end, a suitable notice shall be affixed to the remote start equipment.
7. Close the compressor air outlet valve and depressurize the compressor before connecting or disconnecting a pipe.
8. Before removing any pressurized component, effectively isolate the machine from all sources of pressure and relieve the entire system of pressure.
9. Never use flammable solvents or carbon tetrachloride for cleaning parts. Take safety precautions against toxic vapours of cleaning liquids.
10. Scrupulously observe cleanliness during maintenance and repair. Keep dirt away by covering the parts and exposed openings with a clean cloth, paper or tape.
11. Never weld or perform any operation involving heat near the oil system. Oil tanks must be completely purged, e.g. by steam cleaning, before carrying out such operations. Never weld on, or in any way modify, pressure vessels.
12. Whenever there is an indication or any suspicion that an internal part of a machine is overheated, the machine shall be stopped but no inspection covers shall be opened before sufficient cooling time has elapsed; this to avoid the risk of spontaneous ignition of the oil vapour when air is admitted.

13. Never use a light source with open flame for inspecting the interior of a machine, pressure vessel, etc.
14. Make sure that no tools, loose parts or rags are left in or on the machine.
15. All regulating and safety devices shall be maintained with due care to ensure that they function properly.
They may not be put out of action.
16. Before clearing the machine for use after maintenance or overhaul, check that operating pressures, temperatures and time settings are correct. Check that all control and shut-down devices are fitted and that they function correctly. If removed, check that the coupling guard of the compressor drive shaft has been reinstalled.
17. Every time the separator element is renewed, examine the discharge pipe and the inside of the oil separator vessel for carbon deposits; if excessive, the deposits should be removed.
18. Protect the motor, air filter, electrical and regulating components, etc. to prevent moisture from entering them, e.g. when steam cleaning.
19. Make sure that all sound-damping material and vibration dampers, e.g. damping material on the bodywork and in the air inlet and outlet systems of the compressor, is in good condition. If damaged, replace it by genuine material from the manufacturer to prevent the sound pressure level from increasing.
20. Never use caustic solvents which can damage materials of the air net, e.g. polycarbonate bowls.
21. **The following safety precautions are stressed when handling refrigerant:**
 - Never inhale refrigerant vapours. Check that the working area is adequately ventilated; if required, use breathing protection.
 - Always wear special gloves. In case of refrigerant contact with the skin, rinse the skin with water. If liquid refrigerant contacts the skin through clothing, never tear off or remove the latter; flush abundantly with fresh water over the clothing until all refrigerant is flushed away; then seek medical first aid.



Also consult following safety precautions: [Safety precautions during installation](#) and [Safety precautions during operation](#).

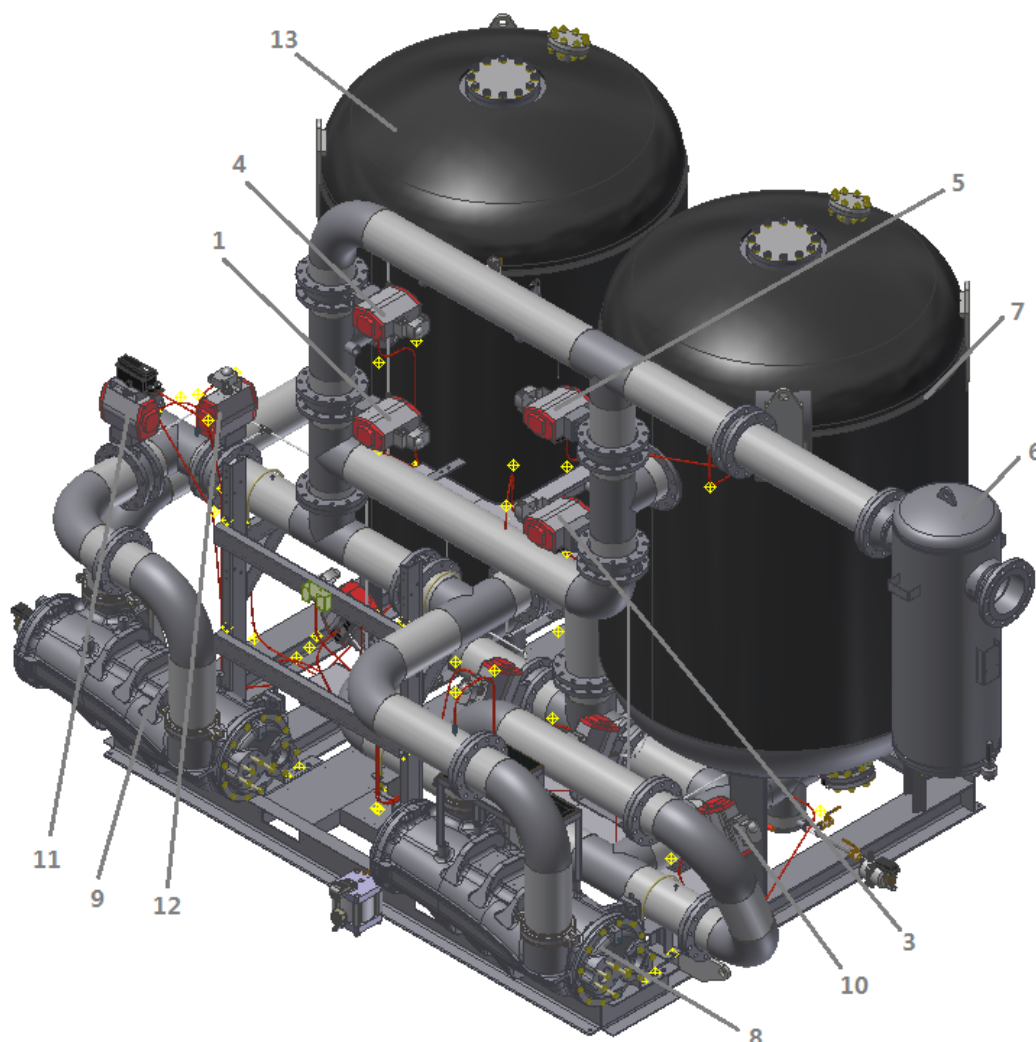
These precautions apply to machinery processing or consuming air or inert gas. Processing of any other gas requires additional safety precautions typical to the application which are not included herein.

2 GENERAL DESCRIPTION

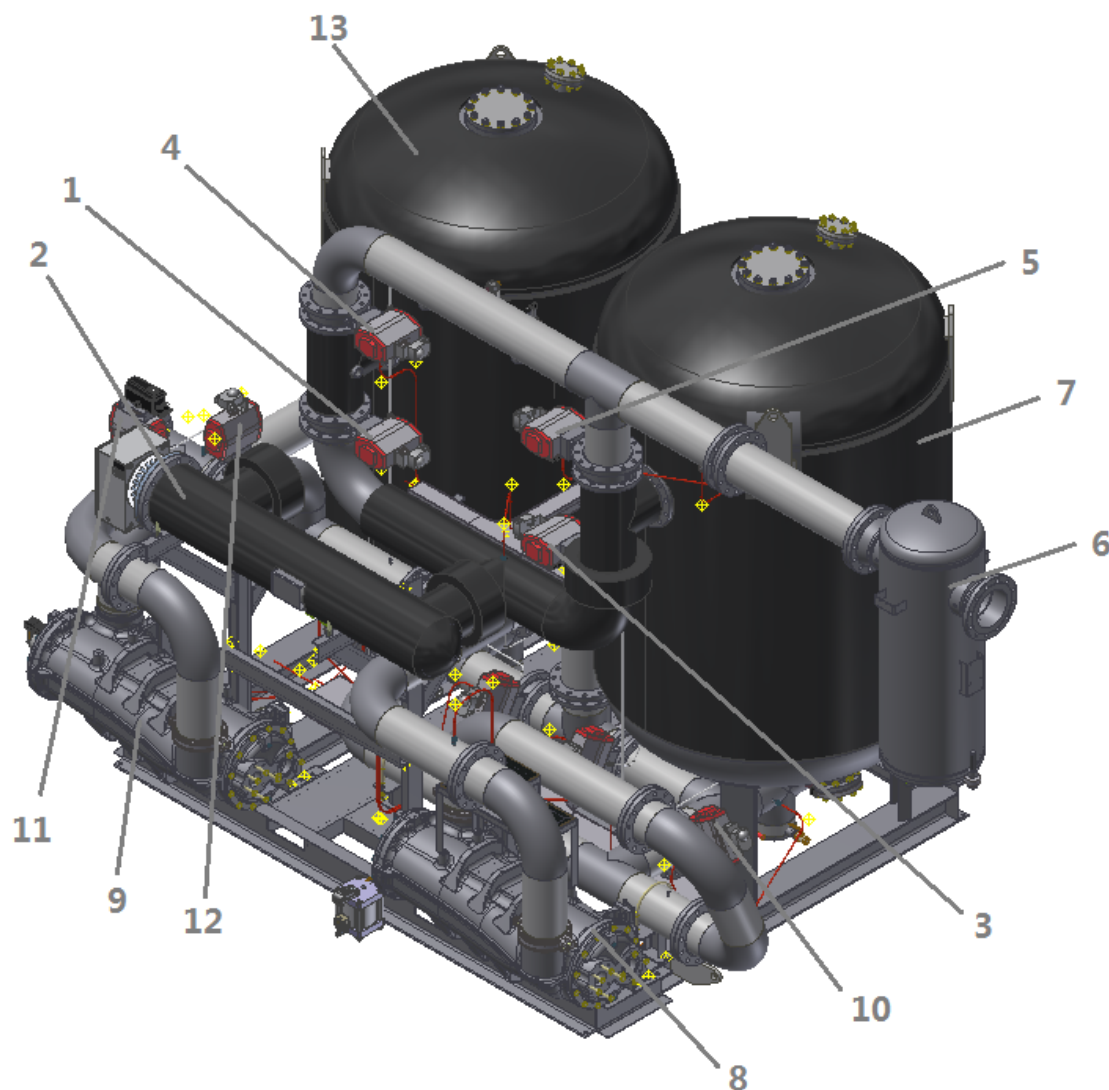
2.1 INTRODUCTION

DESCRIPTION

The XD air dryers remove moisture from compressed air by adsorption without air loss. The air dryer consists of two drying towers (vessels) containing desiccant. While one tower adsorbs moisture, the other is regenerating (or waiting in standby after regeneration). In general the tower function reverses every 4 to 8 hours, but a pressure dew point meter can extend the shift time of the towers up to 24 hours to gain even more energy savings.



General view, typical XD-S-type dryer (Zero Purge-cooled)



General view, typical XD-G-type dryer (Zero Purge-cooled)

Reference	Description
1	Inlet valve of regeneration circuit (V2)
2	Heater
3	Inlet valve of regeneration circuit (V1)
4	Outlet valve (V3)
5	Dry air outlet
6	Filter (optional)
7	Adsorption vessel (tower A)
8	Cooler
9	Cooler air valve (V16)
10	Bypass valve of regeneration circuit
11	Cold air inlet
12	Cooler air valve (V17)

Reference	Description
13	Hot air inlet
14	Inlet valve of hot air circuit (V30)
15	Safety valve
16	Adsorption vessel (tower B)
17	Outlet valve (V4)
18	Safety valve

The dryer can produce pressure dew points ranging from 10 °C (50 °F) down to -40 °C (-94 °F). Once the desiccant is saturated, it can be regenerated at temperatures between 90 °C (194 °F) and 200 °C (392 °F).

The even round shape of the beads and their vitreous, glossy surface ensure an evenly distributed flow, low pressure drops, low friction and a very low dust emission.

If the dryer is maintained as described further and it is operated in normal operation conditions, the drying beads can adsorb moisture and regenerate time after time up to 5 years of continuous operation.

Alarm functions are included for electronic water drain failure, low working pressure, shifting failure, valve failure, sensor error, etc...

The sonic nozzle (optional) should be installed downstream of the dryer and avoids high air flow velocities.

S AND G-TYPE DRYERS

An XD dryer is available in 2 types:

- The S or Suppression type XD dryer has no heater in the regeneration air circuit. This allows dew point variation in certain circumstances.
- The G or Guaranteed type XD dryer has a thermostat controlled heater in the regeneration air circuit.

The heater guarantees the dew point in all conditions and is disabled by a number of alarm routines to avoid the production of excessive heat.

SECOND COOLER FOR ZERO PURGE-COOLED VERSIONS

A Zero Purge-cooled dryer is provided with two coolers one of which is only used for the cooling stage. In regeneration and standby time this cooler is used to lower the adsorption air temperature. The second cooler improves Pressure Dew point (PDP) and extends the cycle time, which will lower the average energy consumption.

ELECTRONIC CONDENSATE DRAIN TRAPS

Electronic drain traps are provided on vessels and cooler(s). They include the electronic drain power supply and the high level condensate detection in case there is drain trap malfunction. Manual drains are provided as backup and for servicing.

DRYER UNLOAD DETECTION

An XD dryer comes with a dryer unload detection based on a pressure difference measurement. When the air flow stops, the pressure drop sensor measures a sudden, lower pressure difference. This stops the dryer function, enables power saving and ensures the dryer safety and protection. The detection enables next functionalities:

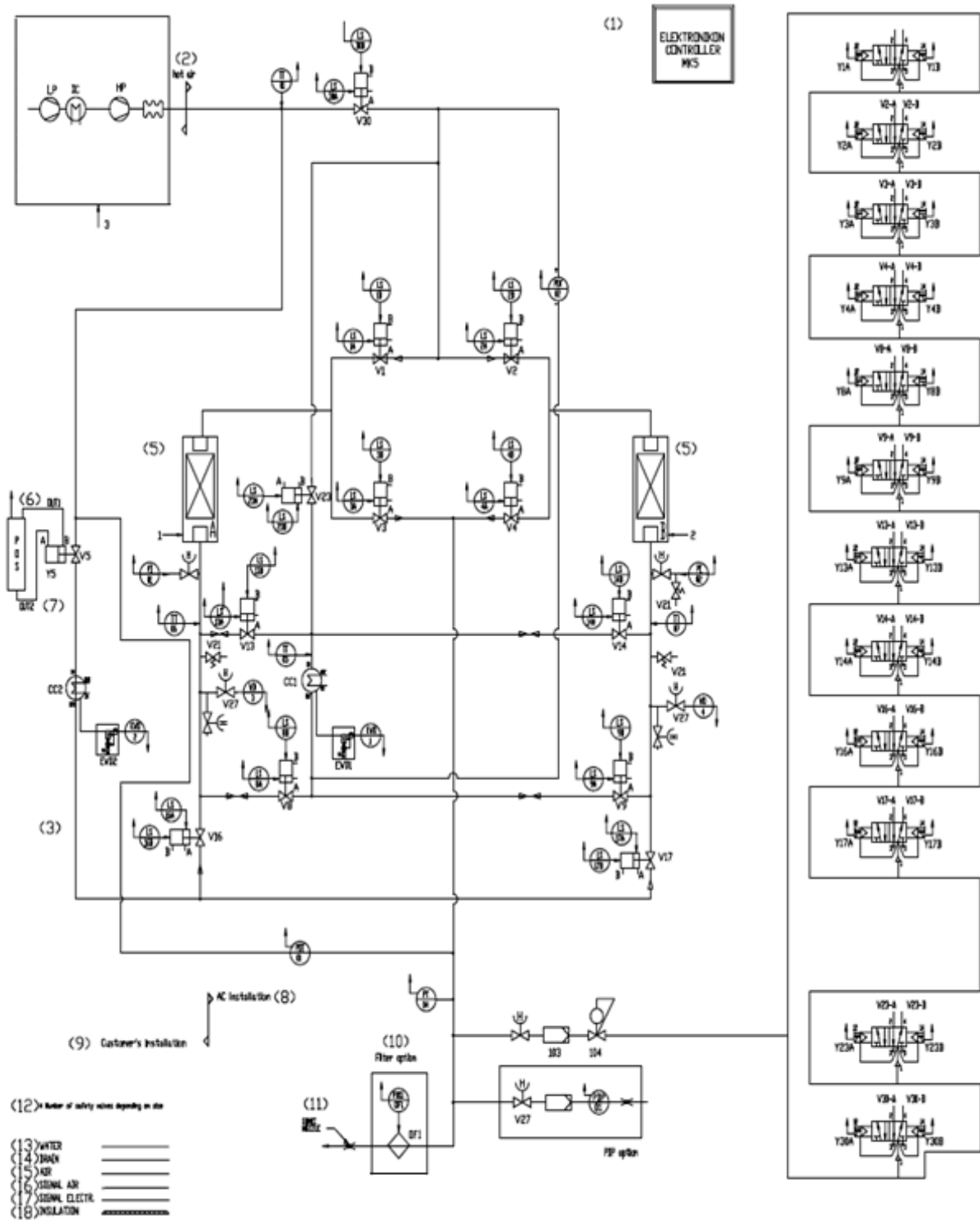
- Pausing the (vessel shift) timers.
- Cutting the power to the heater. (of G-type XD dryers)

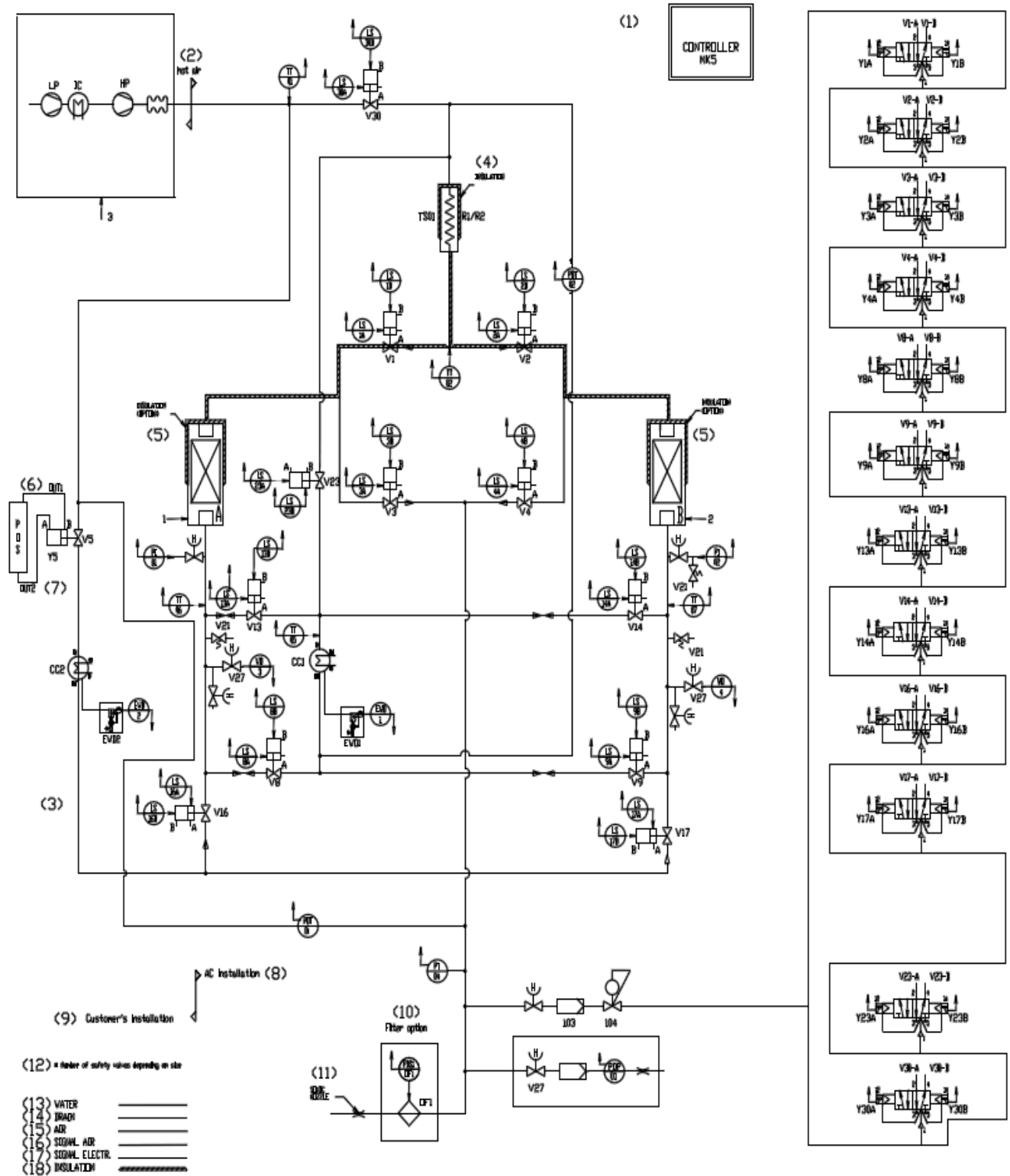
SMARTBOX

An XD dryer comes with a mounted **SMARTBOX** in the electrical cabinet. The **SMARTBOX** allows read-out of the dryer parameters on a login-protected web site called **SMARTLINK**. The connected antenna is mounted on the outside of the electrical cabinet. (Refer to the **SMARTBOX** and **SMARTLINK** section.)

2.2 FLOW DIAGRAM

PIPING & INSTRUMENTATION DIAGRAM





Flow diagram, XD G-type dryer

Reference	Description
Solenoid Name A	Valve open
Solenoid Name B	Valve closed
CC1	Cooler
CC2	Cooler

Reference	Description
EWD1	Electronic water drain CC1
HP	High pressure
LP	Low pressure
LS	Limit switch
OF1	Outlet filter (option)
PDP01	Pressure dew point
PDS1	Pressure difference over outlet filter
PDT01	Pressure difference over dryer
PI	Pressure indicator
POS	Analog positioner
PT04	Pressure outlet
R1/R2	Heater elements
TS01	Overheating switch
TT01	Temperature, hot air inlet
TT02	Temperature, heater outlet
TT05	Temperature, cooler inlet
TT06	Temperature, bottom of vessel A
TT07	Temperature, bottom of vessel B
V1	Inlet valve of regeneration circuit
V13	Valve A of regeneration circuit
V14	Valve B of regeneration circuit
V16,17	Cooler valves
V2	Inlet valve of regeneration circuit
V20	Safety valve (full flow)
V21	Safety valves on towers (optional)
V23	Cooler valve
V25	Check valve of hot air inlet
V27	Manual valve
V3, V4	Outlet valves
V30	Inlet valve of hot air circuit
V5	Bypass valve of regeneration circuit
V8, V9	Outlet valves of cooler
1	Vessel A
2	Vessel B
3	Compressor
101	Dust filter
103	Pilot air filter

Reference	Description
105	Bi-stable 5/2 spool valve
112	Flow restriction, single direction
(1)	Elektronikon controller Mk5
(2)	Hot air
(3)	Cold air
(4)	Insulation
(5)	Insulation (option)
(6)	Out 1
(7)	Out 2
(8)	Atlas Copco Installation
(9)	Customer's installation
(10)	Filter option
(11)	Sonic option
(12)	Notes: number of safety valves depending on size
(13)	Water
(14)	Drain
(15)	Air
(16)	Signal air
(17)	Signal electric
(18)	Insulation

DESCRIPTION

The hot compressed air can enter the system in three ways:

1. Hot wet compressed air enters the system via the regeneration by-pass valve directly to the cooler(s) to cool the air.
2. Hot wet compressed air enters the system via the regeneration inlet valve and goes through the side inlet of the vessel (A or B) for regeneration. The air leaves at the bottom of the vessel and goes through the check valve to the cooler(s).
3. A combination of 1 and 2 for regeneration with heaters (Guaranteed dew point (G-Type)).

After passing the cooler stage, the cooled air flows to the bottom of one of the towers (A or B) and passes the desiccant for adsorption of the moisture. The dry compressed air leaves the dryer via the outlet valve.

2.3 REGENERATION STEP

DESCRIPTION, ZERO PURGE-COOLED VERSIONS

Regeneration

The moist desiccant beads are dried during the regeneration step. The hot air from the compressor is sent through the regeneration inlet valve through the vessel A or B from top to bottom before it enters the combicooler stages.

For the dew point Suppression (S-) type: Sensors on top of the vessels monitor the regeneration temperatures. If the temperature of the bottom of a vessel reaches the preset value, the cooling step will be started.

Regeneration with heaters (only with G-type)

For the Guaranteed dew point (G-) type, an extra heating step (regeneration with heaters) is performed following the regeneration step. In this extra step a part of the compressor flow will be heated to a fixed temperature using electric heaters and a flow control valve, forcing more moisture out of the vessel and making the top part of the desiccant bed extra dry. The rest of the compressed air flow will go directly to the combicooler stage.

For the Guaranteed dew point (G-) type: Sensors on top of the vessels monitor the regeneration temperatures. If the temperature difference between the top and bottom of a vessel reaches the preset value, the cooling step will be started.

It is possible to convert a G-type into an S-type by adjusting a setting in the Elektronikon regulator. Refer to the Elektronikon regulator section.

2.4 COOLING STEP

DESCRIPTION

After the full-flow regeneration without heaters or the regeneration with heaters (on G-type) the cooling step starts. This step is required to avoid high air temperatures of the air leaving the dryer.

The dryer is supplied with the Zero Purge cooling mode. This implies that the hot air from the compressor is sent through the first of the two combicoolers via the regeneration by-pass valve. Then the air flows upward through vessel A or B via the bottom valve, thereby cooling the desiccant quickly, but also pre-loading the bottom of the desiccant column with water of the wet cooling air. The air leaves the vessel and goes through the wet air cooling valve to the second cooler for adsorption. The air then enters vessel A or B and leaves the dryer through the outlet valve.

2.5 STANDBY STEP

DESCRIPTION

Once the regenerated column is cooled down, the cooling flow stops to save energy. The switching point is determined by three conditions:

- The minimum cooling time of the regenerated vessel is 30 minutes.
- The maximum cooling time of the regenerated vessel is 60 minutes.
- The temperature difference over the vessel is 20 °C (36 °F).

2.6 SHIFTING TOWERS

DESCRIPTION

Once the dew point in the used dryer vessel exceeds -40 °C (-40 °F) (refer to the Elektronikon regulator section for range adaptation), the dryer vessels (towers) shift function:

- The standby tower becomes the used tower.
- The previously used tower starts regenerating.

Shifting of the towers happens without causing a pressure drop in the air net.

The compressed air entering the dryer combicooler stage now flows to the other tower. Immediately after the valves have changed position, the regeneration inlet valve will open to start regeneration. Check the *Regeneration step* section.



The dryer can only shift towers:

- When the pressure difference between both vessels A and B is less than 1 bar (14.5 psi).
- No valve alarms occur.

2.7 CYCLE TIME OVERVIEW

OVERVIEW

The dryer sequence is as follows:

XD	Version	Regeneration (Minutes)	Regeneration with heaters
S	Zero Purge-cooled	90-150	--
G	Zero Purge-cooled	60-120	60-120

XD	Version	Cooling, Zero Purge (Minutes)	Standby (Hours)	Total cycle time (Hours)
S	Zero Purge-cooled	30-60	0-21	4-24
G	Zero Purge-cooled	30-60	0-21	4-24

The regulator stops counting the operating cycle time if a compressor unload is detected by a digital contact.

TOTAL CYCLE TIME

Depending on the PDP control setting (refer to the Elektronikon regulator section) the cycle time may differ:

- PDP control = OFF

The minimum cycle time is 4 hours.

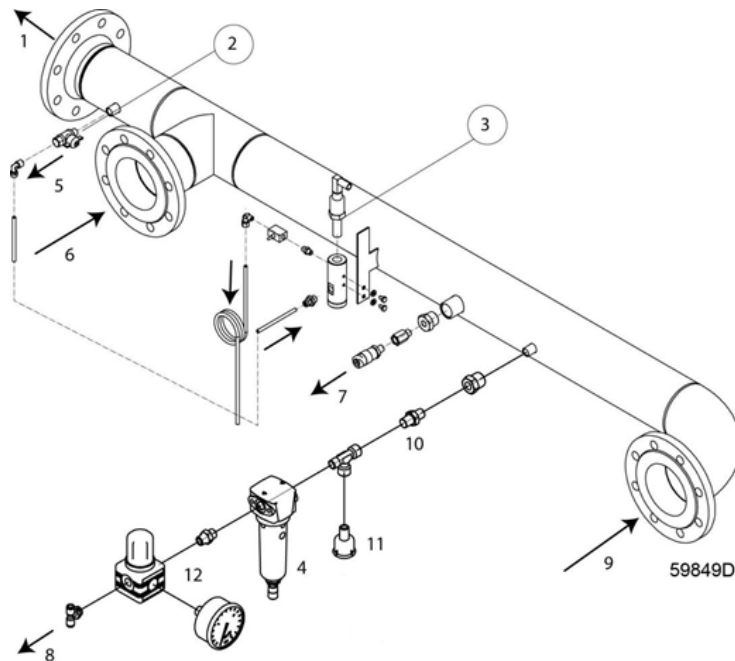
- PDP control = ON

The minimum cycle time is 3 hours.

The installed dew point sensor can extend the adsorption cycle up to 24 hours if there is still adsorption capacity left. This measurement saves considerable amounts of energy.

2.8 PNEUMATIC ACTUATION OF BUTTERFLY VALVES

DRAWING



Exploded view on the air outlet pipe of the dryer.

Drawing Reference	Description
1	Flow to dryer outlet
2	Valve to connect pressure dew point sensor
3	Pressure dew point sensor with integrated filter
4	Pilot air filter
5	Flow to pressure dew point sensor
6	Flow from air dryer vessel / tower A
7	Flow to outlet pressure sensor
8	Flow to pneumatic actuators of butterfly valves
9	Flow from air dryer vessel / tower B
10	Blind hexagon nut
11	Socket
12	Pressure regulator with manometer

DESCRIPTION

Pilot air is used to supply air to the pneumatic actuators of the butterfly valves. Tuning of the pilot air pressure is not required as the pressure range of the actuators equals the working pressure of the dryer. A pilot air filter is mounted to avoid desiccant (dust) from entering the actuators.

In case the working pressure is below 4 bar (58 psi), the external pilot air kit should be mounted (refer to the *Options* section).

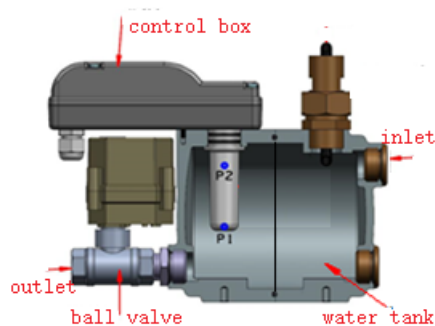
2.9 CONDENSATE DRAIN SYSTEM

DESCRIPTION

The condensate enters the Electronic Water Drain (EWD) via inlet and accumulates in the water tank. A capacitive sensor continuously measures the liquid level. As soon as the collector is filled up to a certain level, a drain delay timer is started. As soon as the delay time has expired, ball valve is activated and diaphragm opens outlet, discharging the condensate.

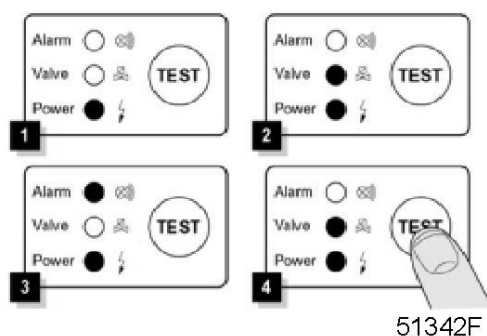
When the tank has been emptied, the outlet closes quickly without wasting compressed air.

When the controller registers a malfunction, the red alarm LED starts flashing, a warning appears on the display of the regulator and the electronic drain valve will automatically change to the alarm mode, opening and closing the valve according to a sequence as shown below. This condition continues until the fault is remedied. If the fault is not remedied automatically, maintenance is required.



Electronic water drain

TESTING THE ELECTRONIC WATER DRAIN



51342F

Control panel, EWD

Briefly press the TEST button and check that the valve opens for condensate discharge.

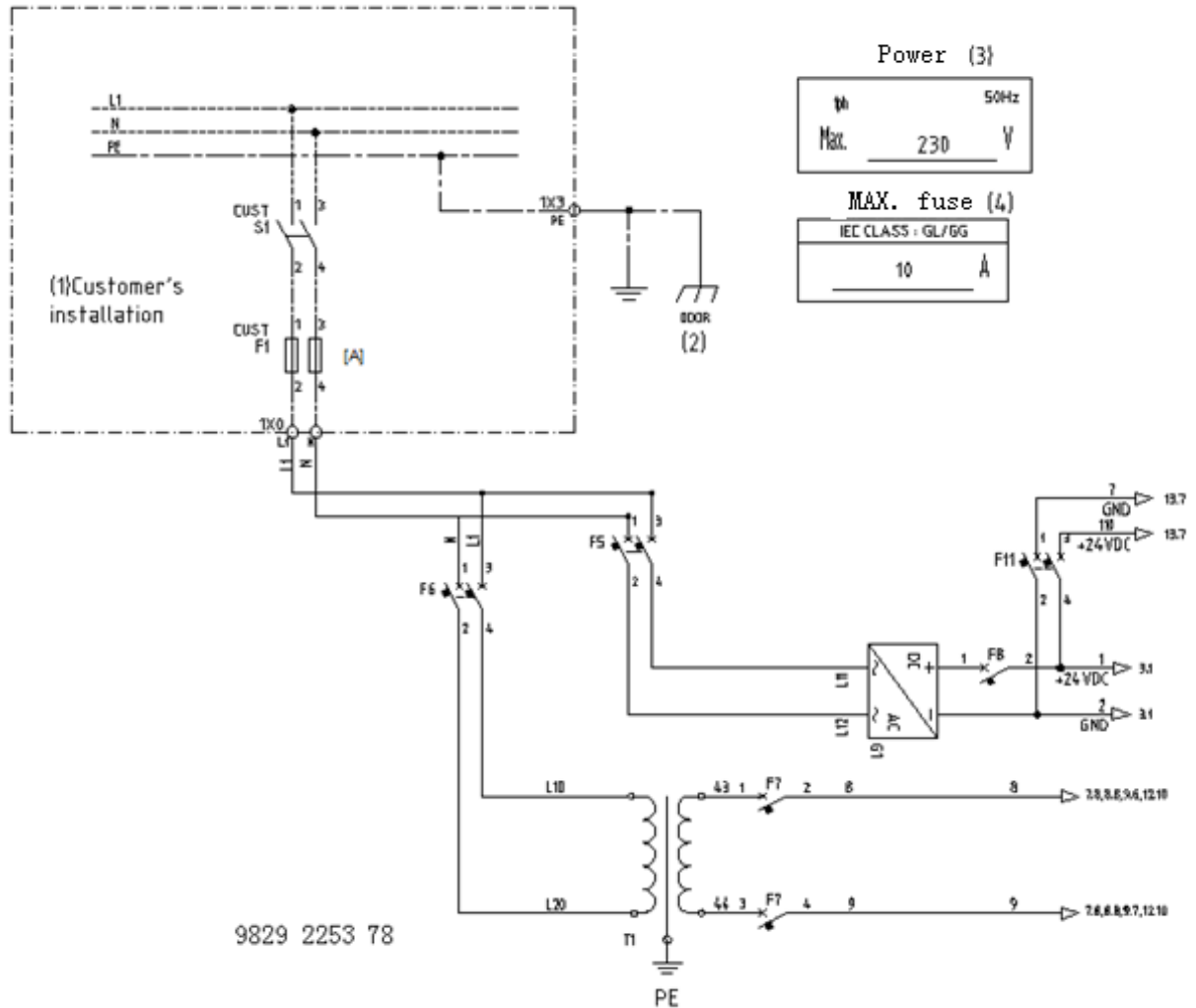
Checking the alarm signal:

- Press and hold the test button for at least 1 minute.
- Check that the alarm LED flashes.
- Check that a warning is generated on the display of the regulator.

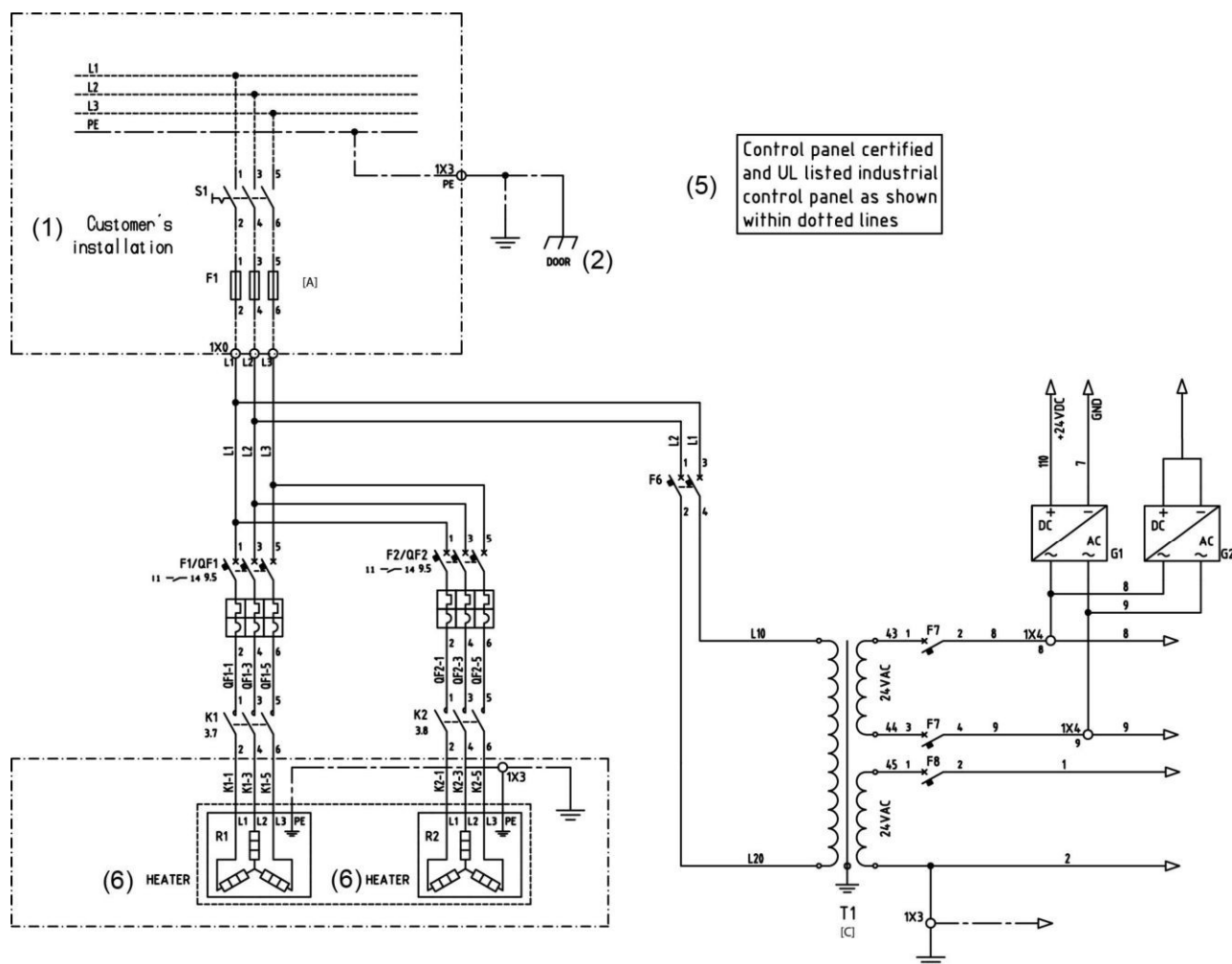
- Release the test button.

2.10 ELECTRICAL SYSTEM

ELECTRICAL DIAGRAM



Electrical diagram for S-type



9827 2475 00-01
59822D

Electrical diagram for G-type

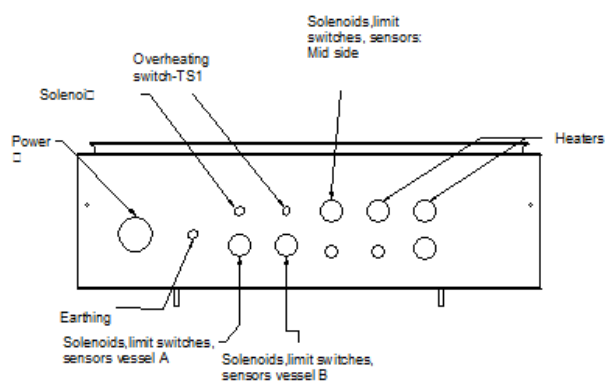
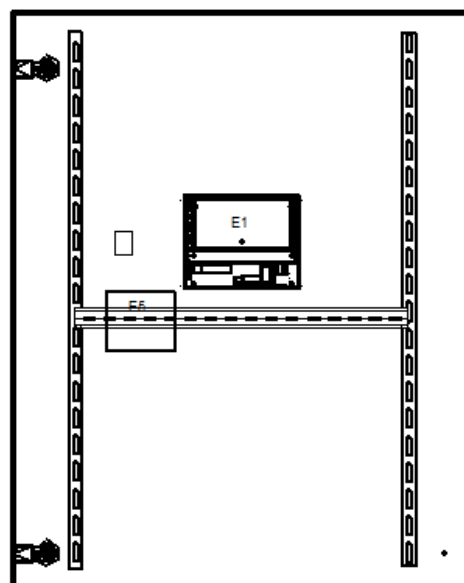
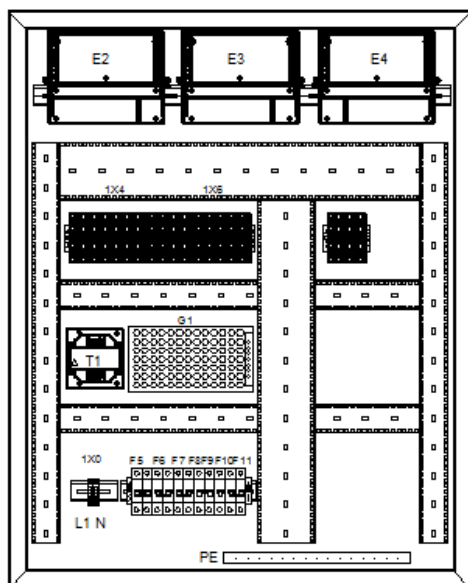
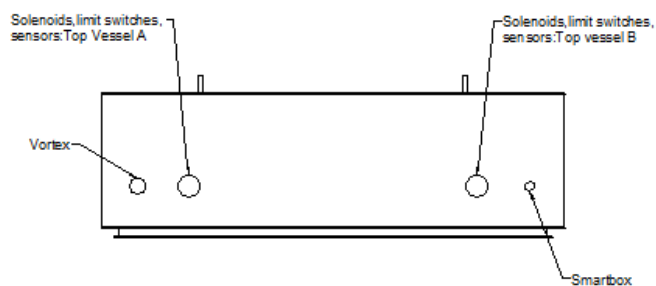
Drawing reference	Description
(1)	Customer's installation
(2)	Door
(3)	Supply
(4)	Maximum fuse
(5)	Certified and UL listed industrial control panel as shown within dotted lines
(6)	Heater
[A]	Maximum fuses with regard to short circuit protection of cubicle. Cable section may impose a fuse of smaller value.

Drawing reference	Description
[B]	Tightening torque of bolts: - for contactors - metric thread size M6 8 Nm (6 lbft) - metric thread size M8 20 Nm (15 lbft) - metric thread size M10 40 Nm (29 lbft) - for terminals
[C]	To connect correct transformer voltage, check the data plate of the transformer for correct connections conform the supply voltage.

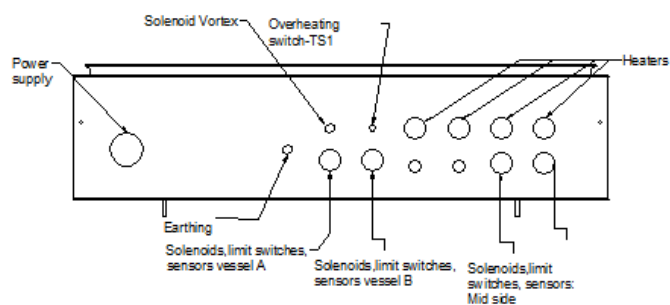
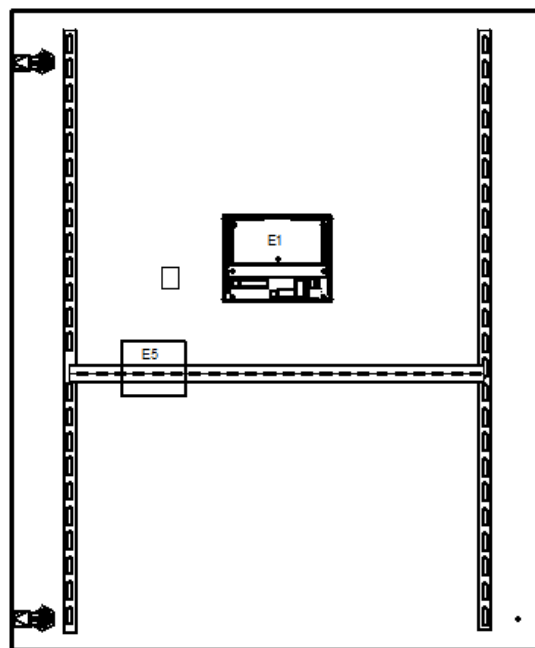
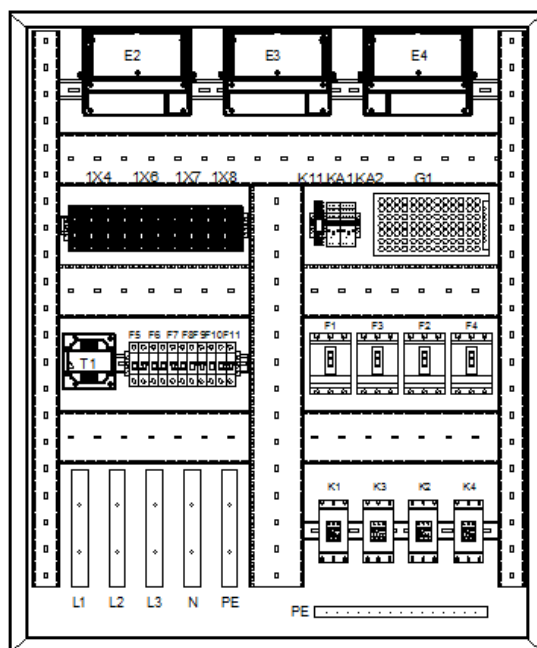
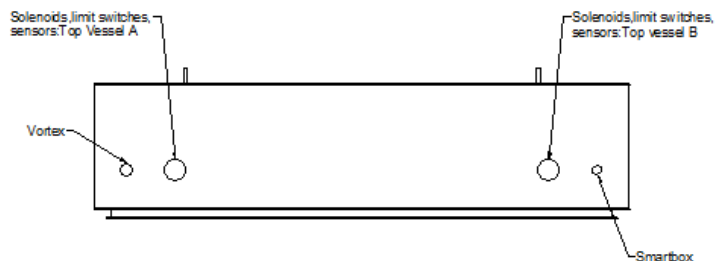
NOTE

	The entire electrical diagram is included in the delivery. (Check the electrical cabinet door)
---	--

POSITION OF ELECTRICAL COMPONENTS



View on the electrical cabinet XD S



View on the electrical cabinet XD G

Drawing reference	Description
E1	Elektronikon regulator
E2	Extension module of the Elektronikon regulator: IO2 Digital-analog In + Out
E3	Extension module of the Elektronikon regulator: IO2 Digital-analog In + Out
E4	Extension module of the Elektronikon regulator: IO34 Analog Out

Drawing reference	Description
E5	Smartbox (to check the Elektronikon read-out on a website) (Refer to the SMARTBOX and SMARTLINK section)
F1	Fuse of main heater circuit
F2	Fuse of regulating heater circuit
F5	Fuse of transformer
F6	Fuse of transformer T1
F7	Fuses of Elektronikon regulator
F8	Fuse of control circuit
F9	Fuse of electronic condensate drain EWD1,3
F10	Fuse of electronic condensate drain EWD2,4
G1	AC/DC converter
K1 K3	Contactor of main heater circuit
K2 K4	Contactor of regulating heater circuit
K11	Contactor of overheating switch
B1	Optocoupler for 4/20 mA dew point read-out (optional)
S3	Emergency stop
T1	Transformer
TS1	Overheating switch
L1 L2 L3 N PE	Terminal
1X4	Terminal for 24VAC connections
1X6	Terminal for Elektronikon regulator connections
1X7	Terminal for potential free contacts of the customer
1X8	Terminal for optional inputs

2.11 EXTERNAL INDICATION OF DRYER STATUS

AUXILIARY CONTACTS

The dryer is standard equipped with 5 potential-free contacts:

- 1X7-K01: Dryer started (contact closed = dryer started)
- 1X7-K02: General warning (contact closed = no alarm)
- 1X7-K03: General shut-down (contact closed = no shut-down active)
- 1X7-K04: Pressure dew point alarm (contact closed = no alarm)
- 1X7-K05: Dryer loaded (contact closed = dryer not loaded)

As optional equipment, a second pressure dew point readout is available as an external analog 4-20 mA signal (using an optocoupler). Terminals 5 and 6 on the module should be used.

Indication	Relay	Terminals on strip 1X7	Maximum load
Dryer started	K01	100-101	10 A / 250V AC
General warning	K02	102-103	10 A / 250V AC
General shut-down	K03	104-105	10 A / 250V AC
Pressure dew point	K04	106-107	10 A / 250V AC
Dryer loaded	K05	108-109	10 A / 250V AC

The contact characteristics (conforming to IEC 947-5-1) of the electromechanical relay modules are as follows:

- Rated operating voltage: up to 250V AC / 125V DC
- Rated operating current per 1 million operating cycles

Utilization category	Rated operating current (A)
AC-12 230V AC	4
AC-13 230V AC	1
AC-14 230V AC	1
AC-15 230V AC	1
DC-12 24V DC	5
DC-13 24V DC	1

WARNING



Stop the dryer and switch off the voltage before connecting external equipment.
Consult your nearest Atlas Copco Customer Center.

2.12 DRYER STARTING ROUTINE

DESCRIPTION

The dryer starting routine prevents the dryer from starting with wrong valve positions or incorrect vessel pressure when going back to its normal regeneration routine. This routine also prevents blocking of the compressed air flow by the dryer.

The dryer starting routine depends on the way the compressor was stopped. 3 possible scenarios can occur:

1. Restart after manual stop, remote stop or alarm shutdown: The dryer cycle continues where it was interrupted.
2. Restart after power failure: The dryer cycle continues where it was interrupted once the power supply has returned for 5 minutes.
3. Restart after a change of the parameter set: of the Elektronikon regulator. The dryer cycle starts with the standby step of tower B. Refer to the *Standby step* section.

IMPORTANT



The dryer starting routine cannot prevent high flow velocities in the towers during start-up. These can be harmful to the desiccant and should be avoided by carefully

2.13 SMARTBOX AND SMARTLINK

GENERAL

An XD dryer comes with a **SMARTBOX** mounted in the electrical cabinet.



59850F

The **SMARTBOX** allows read-out of the dryer parameters on a user login-protected web site called **SMARTLINK**. The connected antenna is mounted on the outside of the electrical cabinet.

SAFETY PRECAUTIONS

It is important to follow all regulations regarding the use of radio equipment, in particular regarding the possibility of radio frequency (RF) interference. Please follow the safety advice given below carefully.

- Respect restrictions on the use of radio equipment in fuel depots, chemical plants or other explosive environments.
- Avoid **SMARTBOX** operation close to adequately protected personal medical devices such as hearing aids and pacemakers. Consult the manufacturers of the medical device to determine if it is adequately protected.
- Avoid **SMARTBOX** operation close to other electronic equipment, which may also cause interference if the equipment is inadequately protected. Observe any warning signs and manufacturer recommendations.
- Respect a distance from the human body of at least 20 cm (8 inch) during **SMARTBOX** operation.
- Do not operate the **SMARTBOX** in areas where cellular modems are not advised without proper device certifications. These areas include environments where cellular radio can interfere, such as explosive atmospheres, medical equipment, or any other equipment, which may be susceptible to any form of radio interference. The modem can transmit signals that could interfere with this equipment.

SMARTLINK

The XD dryer is **SMARTLINK**-ready.

SMARTLINK is Atlas Copco's web-based platform to monitor compressed air installations. It is available in 3 product levels:

- **SMARTLINK Service**

SMARTLINK Service rules out all uncertainties. Scheduling maintenance visits becomes as simple and easy as it should be; the service log book is always just one click away and the online link with Atlas Copco allows to request and quickly receive quotes for spare parts or additional services.

This product level is free of charge and available for 3 years following the machine purchase.

- **SMARTLINK Uptime**

SMARTLINK Uptime keeps your compressors up and running. By e-mail and/or text, you receive all relevant machine indications (warnings and shutdowns) in advance. Based on this information, you can then take all necessary actions and measures to avoid the risk of a breakdown.

This product level is available as a free of charge 3-month trial.

- **SMARTLINK Energy**

SMARTLINK Energy safeguards the performance of your equipment. It enables you to continuously monitor and analyze the energy efficiency of the compressor room. You decide which performance

indicators, benchmarks and reports are created. You can make accurate and immediate improvements when needed. The results can be used for energy monitoring according to ISO50001.

This product level is available as a free of charge 3-month trial.

Once a **SMARTLINK** level is chosen, gaining access is the next step. To gain access at least one machine **SMARTLINK**-ready component (compressor, dryer ...) is required. Follow the next steps:

- Gather the information about the **SMARTLINK**-ready machine(s) (e.g. picture of the data plate; invoice...)
- Go to the **SMARTLINK** web site <http://www.atlascopco.com/smartlink>
- Click on the left button in the screen below.



59851F

- Use the data of the **SMARTLINK**-ready machine to register as a user.

SMARTLINK - create account

User details

Language: English

Email: first.last@mycompany.com

Confirm email: first.last@mycompany.com

Time zone: Central European Standard Time

Measurement system: Metric

Company details

Serial number: AIF11111111

Machine name: MyCompressor1

Machine type: ZR400

Year of manufacture: 2012

Customer name: MyCompany

Country: Belgium

Customer center: AC Belgium

Retyp code: c5821f

59821f

Cancel Create

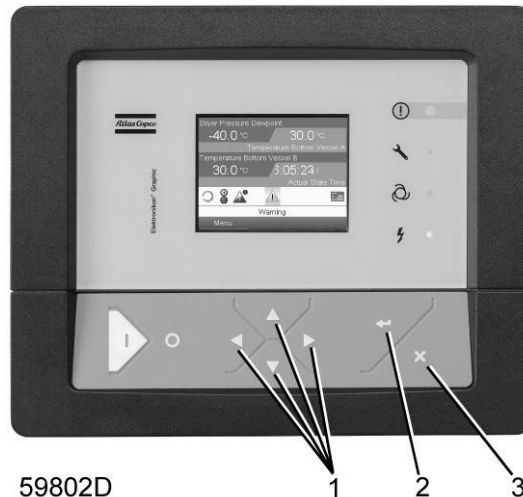
59852F

- You will receive an e-mail with login credentials.
- Go to the **SMARTLINK** web site <http://www.atlascopco.com/smartlink>, log in with the user credentials.
- Enjoy **SMARTLINK**!
- In case more **SMARTLINK**-ready machines are available, you can add these via *My Profile*.

3 ELEKTRONIKON® GRAPHIC CONTROLLER

3.1 ELEKTRONIKON® GRAPHIC CONTROLLER

CONTROL PANEL



59802D

GENERAL DESCRIPTION

The Elektronikon regulator automatically controls and protects the dryer, i.e.:

- keeping the pressure dew-point stable
- monitoring pressures, temperatures and digital switches to ensure safe operation, and stopping the dryer whenever necessary
- restarting the dryer when required

In order to control the dryer and to read and modify programmable parameters, the regulator has a control panel provided with:

- LEDs indicating the status of the dryer
- a display indicating the operating conditions or a fault
- keys to control the dryer and to access the data collected by the regulator
- buttons to manually start and stop the dryer

AUTOMATIC RESTART AFTER VOLTAGE FAILURE

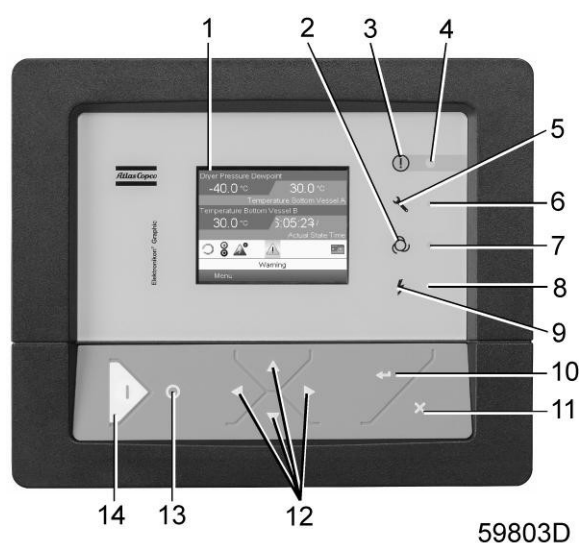
The regulator has a built-in function to automatically restart the dryer if the voltage is restored after voltage failure.



Provided the regulator was in the automatic operation mode, the dryer will automatically restart if the supply voltage to the module is restored within a programmed time period. The power recovery time (the period within which the voltage must be restored to have an automatic restart) can be set between 15 and 3600 seconds or to 'Infinite'. If the power recovery time is set to 'Infinite' the dryer will always

3.2 CONTROL PANEL

ELEKTRONIKON REGULATOR



59803D

PARTS AND FUNCTIONS

Reference	Designation	Function
1	Display	Shows the dryer operating condition and a number of
2	Pictograph	Automatic operation
3	Pictograph	General alarm
4	General alarm LED	Flashes if a shut-down warning condition exists.
5	Pictograph	Service
6	Service LED	Lights up if service is needed
7	Automatic operation LED	Indicates that the regulator is automatically controlling
8	Voltage on LED	Indicates that the voltage is switched on.
9	Pictograph	Voltage on
10	Enter key	Key to activate the selected menu or to modify the
11	Escape key	To go to previous screen or to end the current action
12	Scroll keys	Keys to scroll through the menu.
13	Stop button	Button to stop the dryer. LED (7) goes out.
14	Start button	Button to start the dryer. LED (7) lights up indicating

3.3 ICONS USED

STATUS ICONS

Name	Icon	Description
Stopped/Running	 57786F	When the dryer is stopped, the icon stands still. When the dryer is running, the icon is rotating.
Machine control mode	 57790F	Local start/stop
	 57791F	Remote start/stop
	 57792F	Network control
Automatic restart after voltage failure	 57793F	Automatic restart after voltage failure is active
Week timer	 57794F	Week timer is active
	 57796F	Shutdown
	 57797F	Warning
Service	 57798F	Service required
Main screen	 59162F	Change main screen
	 82196F	Main chart

INPUT ICONS

Icon	Description
	Pressure
	Temperature
	Digital input
	Special protection

SYSTEM ICONS

Icon	Description
	Dryer
	Fan
	Frequency converter
	Drain
	Filter
	Motor
	Failure expansion module
	Network problem
	General alarm

MENU ICONS

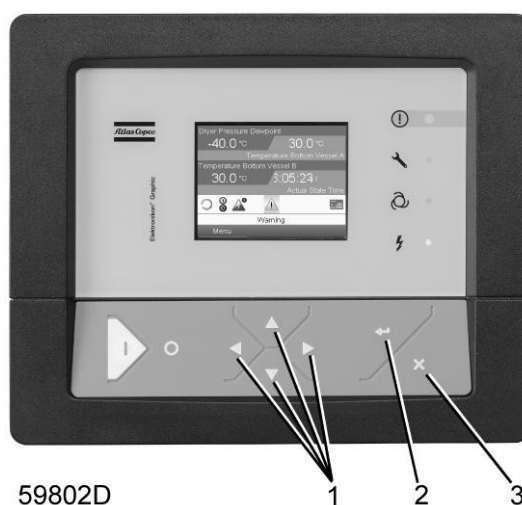
Icon	Description
 57813F	Inputs
 57814F	Outputs
 57812F	Alarms (Warnings, shutdowns)
 57815F	Counters
 58499D	Test
 57817F	Settings
 57798F	Service
 57818F	Event history (saved data)
 57819F	Access key/User password
 57792F	Network
 57820F	Set point
 57867F	Info
 59807D	General settings
 59804D	Valves
 59805D	Heaters
 59806D	Air quality

NAVIGATION ARROWS

Icon	Description
 57821F	Up
 57822F	Down

3.4 MAIN SCREEN

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

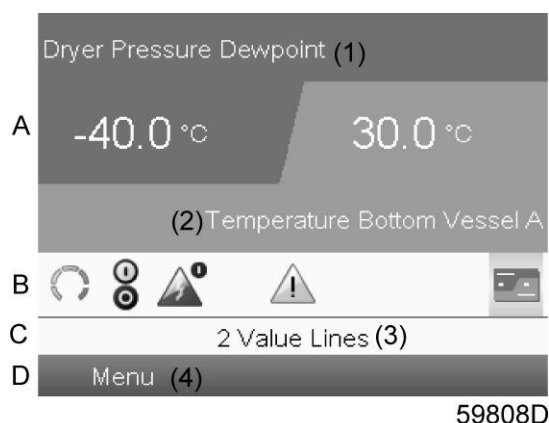
FUNCTION

The Main screen shows the status of the dryer operation and is the gateway to all functions implemented in the controller.

The Main screen is shown automatically when the voltage is switched on and one of the keys is pushed. It is switched off automatically after a few minutes when no keys are pushed.

Five different main screens can be chosen:

1. Two value lines
2. Four value lines
3. Chart (High resolution)
4. Chart (Medium resolution)
5. Chart (Low resolution)



Typical Main screen (Two value lines)

Text on figure

(1)	Dryer, pressure dew point
(2)	Temperature, bottom of vessel A
(3)	2 Value lines
(4)	Menu

- **Section A** shows information regarding the dryer operation (e.g. the Regeneration air inlet, Dryer mix air).
- **Section B** shows Status icons. Following icon types are shown in this field:
 - Fixed icons
 - These icons are always shown in the main screen (e.g. Dryer stopped or running, Dryer status).
 - Optional icons
 - These icons are only shown if their corresponding function is activated (e.g. week timer, automatic restart after voltage failure, etc.).
- Pop up icons

These icons pop up if an abnormal condition occurs (warnings, shutdowns, service,...).

To call up more information about the icons shown, select the icon using the scroll keys and press the enter key.

- **Section C** is called the Status bar.

This bar shows the text that corresponds to the selected icon. When the menu button is selected, this text shows the operational state of the dryer.

- **Section D** shows the Action buttons. These buttons are used:
 - to call-up or program settings.
 - to reset a service message.
 - to have access to all data collected by the regulator.

The function of the buttons depends on the displayed menu. The most common functions are:

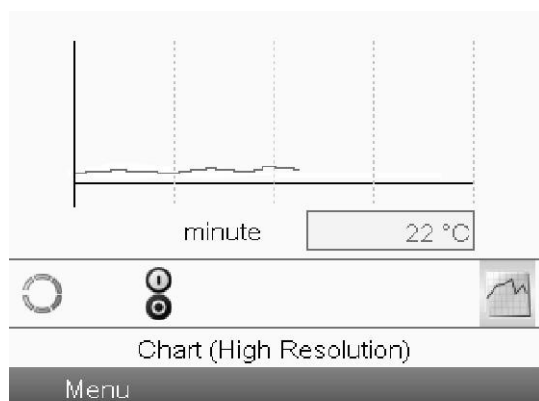
Designation	Function
Menu	To go to the menu

Modify	To modify programmable settings
Reset	To reset a timer or message

To activate an action button, highlight the button by using the Scroll keys and press the Enter key.

To go back to the previous menu, press the Escape key.

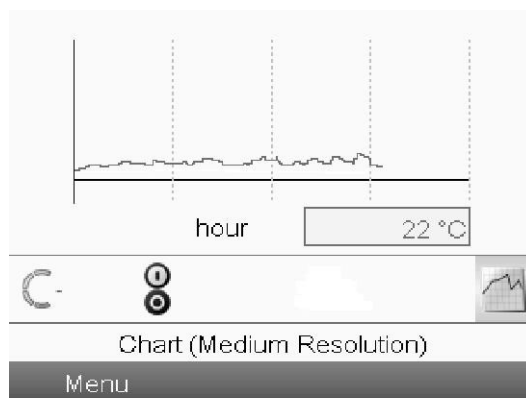
CHART VIEWS



59218D

When the Chart (High Resolution) is selected, a chart showing the temperature per minute is shown on the main screen. The temperature depends on the selection made in the input screen. Each point in the chart is 1 second. The screen shows 4 minutes.

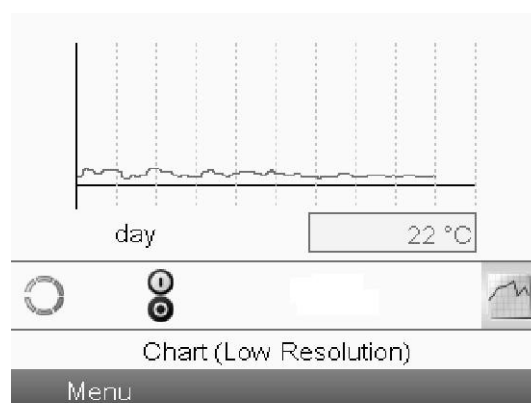
The switch button (icon) for selecting other screens is changed into a small Chart and is highlighted (active).



59219D

When the Chart (Medium Resolution) is selected, a chart showing the temperature per hour is shown on the main screen. The temperature depends on the selection made in the input screen. Each point is the average of 1 minute. The screen shows 4 hours.

The switch button (icon) for selecting other screens is changed into a small Chart and is highlighted (active).



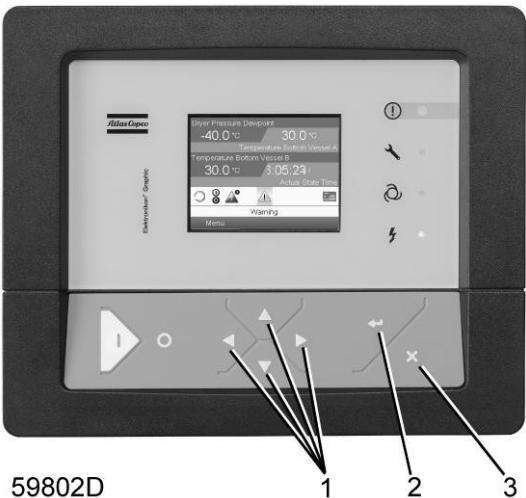
59220D

When the Chart (Low Resolution) is selected, a chart showing the temperature per day is shown on the main screen. This temperature depends on the selection made in the input screen. Each point is the average of 1 hour. The screen shows 10 days.

The switch button (icon) for selecting other screens is changed into a small Chart and is highlighted (active).

3.5 CONTROL MODE SELECTION

CONTROL PANEL



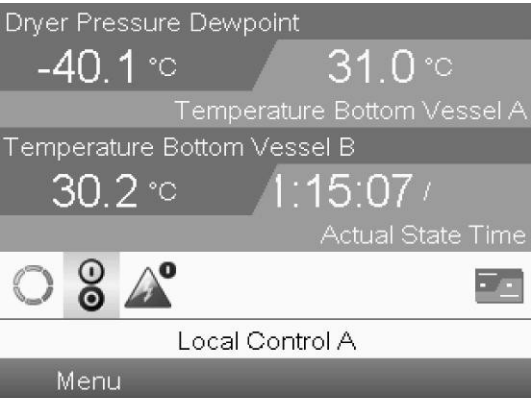
(1)	Scroll keys
(2)	Enter key
(3)	Escape key

FUNCTION

To select the control mode, i.e. whether the compressor is in local control, remote control or controlled via a local area network (LAN).

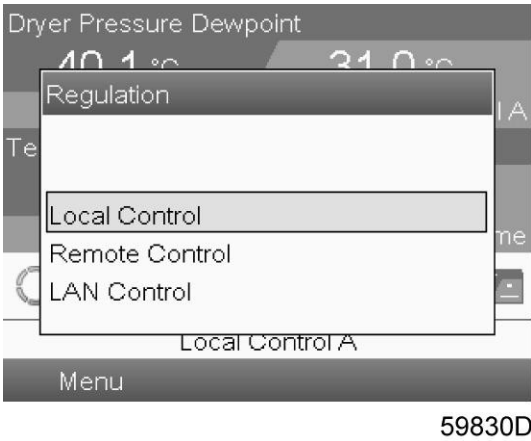
PROCEDURE

Starting from the main screen, use the scroll buttons to go to the Local start/stop icon:



59829D

Press the enter button, following screen appears:



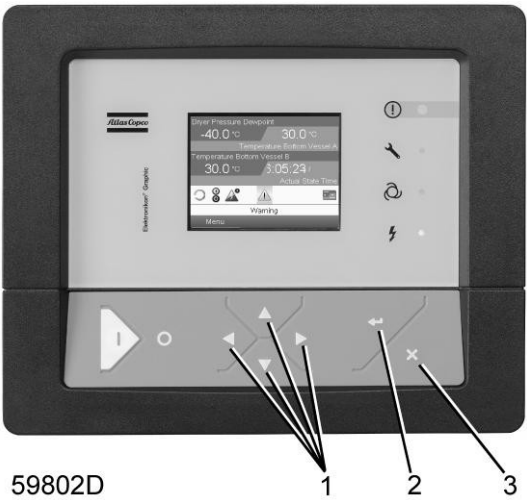
There are 3 possibilities:

- Local control
- Remote control
- LAN (network) control

After selecting the required regulation mode, press the enter button on the controller to confirm your selection. The new setting is now visible on the main screen. See section [Icons used](#) for the meaning of the icons.

3.6 CALLING UP MENUS

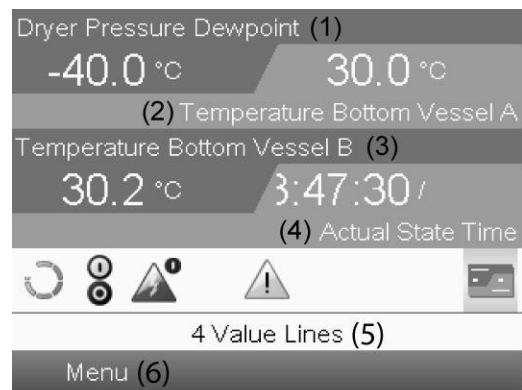
CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

DESCRIPTION

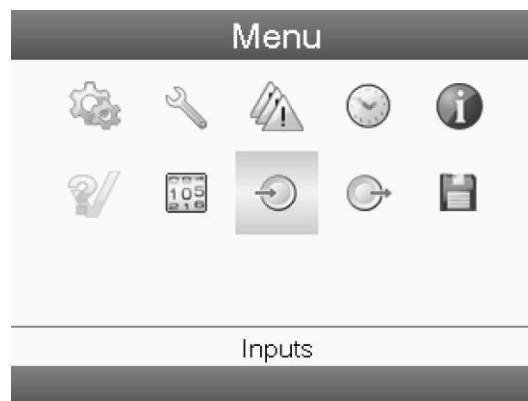
When the voltage is switched on, the main screen is shown automatically (see section [Main screen](#)):



59809D

(1)	Dryer, pressure dew point
(2)	Temperature, bottom of vessel A
(3)	Temperature, bottom of vessel B
(4)	Actual state time
(5)	4 Value Lines
(6)	Menu

- To go to the Menu screen, highlight the Menu button (6), using the Scroll keys.
- Press the Enter key to select the menu. Following screen appears:

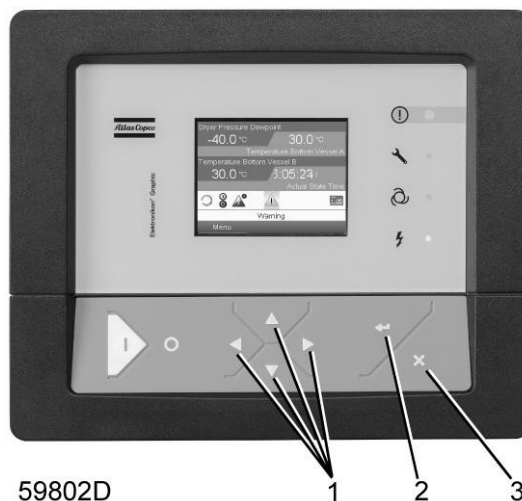


59810D

- The screen shows a number of icons. Each icon indicates a menu item. By default, the Inputs icon is selected. The status bar shows the name of the menu that corresponds with the selected icon.
- Use the Scroll keys to select an icon.
- Press the Enter key (2) to open the menu or press the Escape key (3) to return to the Main screen.

3.7 INPUTS MENU

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, INPUTS



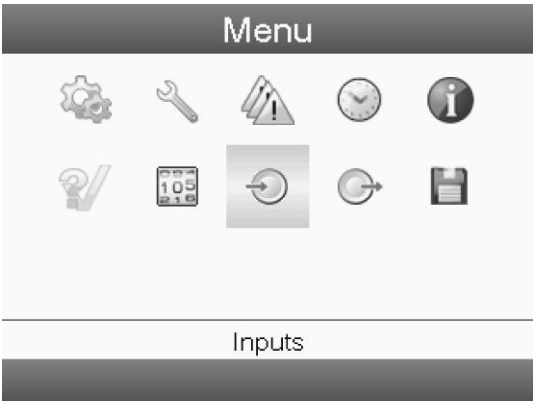
FUNCTION

To call up information regarding the actually measured data and the status of some inputs such as the vessel pressure.

PROCEDURE

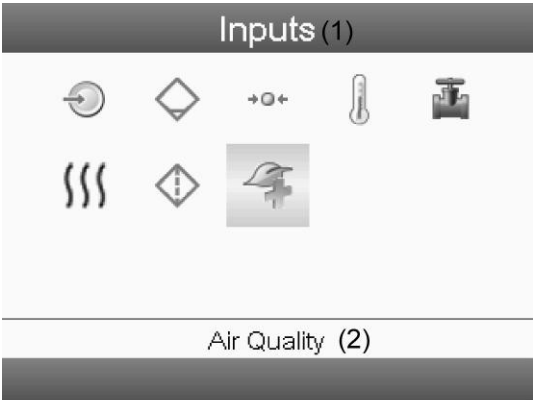
Starting from the main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



59810D

- Press the Enter key. A screen similar to the one below appears:



59815D

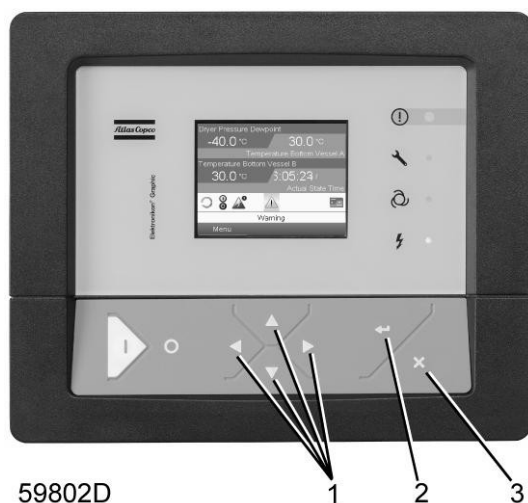
Text on figure

(1)	Inputs
(2)	Air quality

- The screen shows a number of icons. Each icon indicates a menu item. By default, the Air Quality (2) icon is selected. The status bar shows the name of the menu that corresponds with the selected icon.
- Use the Scroll keys to select an icon.
- Press the Enter key (2) to open the menu.
- The screen shows a list of all inputs with their corresponding icons and readings.
- If an input is in warning or shutdown, the original icon is replaced by the warning or shutdown icon respectively.

3.8 OUTPUTS MENU

CONTROL PANEL



59802D

(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, OUTPUTS



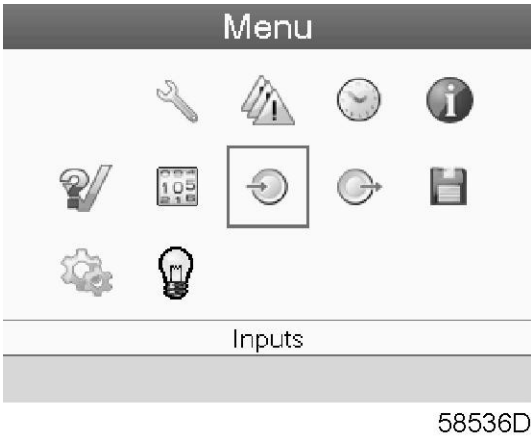
FUNCTION

To call up information regarding the actual status of some outputs such as the Dryer motor, PDP alarm, General shut-down, etc.

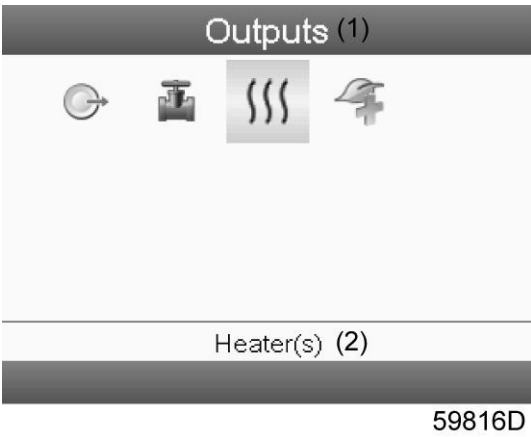
PROCEDURE

Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



- Move the cursor to the Outputs icon using the Scroll keys.
- Press the Enter key. A screen similar to the one below appears:



Text on figure

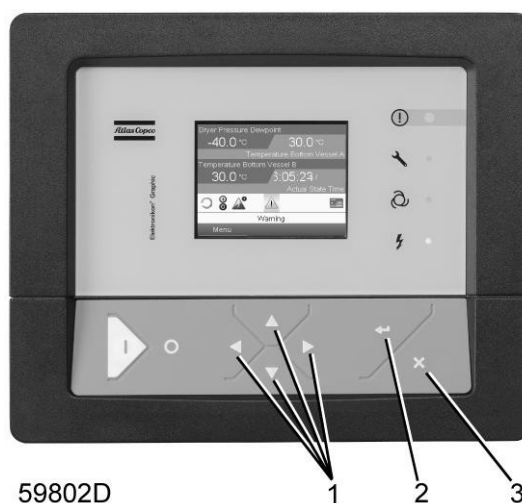
(1)	Outputs
(2)	Heaters

- The screen shows a number of icons. Each icon indicates a menu item. By default, the Heaters (2) icon is selected. The status bar shows the name of the menu that corresponds with the selected icon.

- Use the Scroll keys to select an icon.
- Press the Enter key (2) to open the menu.
- The screen shows a list of all inputs with their corresponding icons and readings.

3.9 COUNTERS

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, COUNTERS



FUNCTION

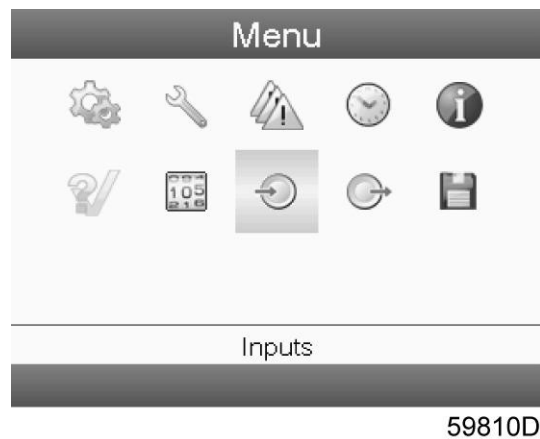
To call up:

- The running hours
- The number of dryer starts
- The number of hours that the regulator has been powered
- The operational state timers

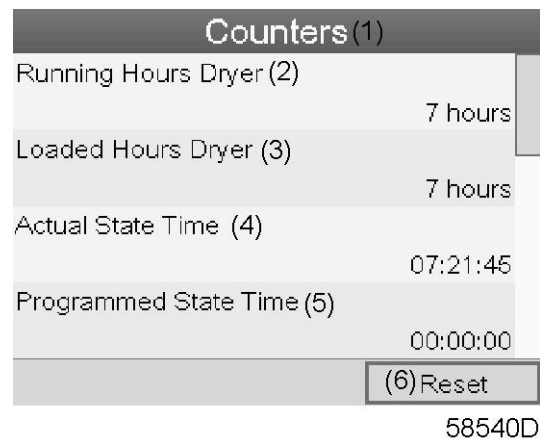
PROCEDURE

Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



- Using the Scroll keys, move the cursor to the Counters icon (see above, section Menu icon).
- Press the Enter key. Following screen appears:



Text on figure

(1)	Counters
(2)	Running hours of dryer
(3)	Loaded hours of dryer
(4)	Actual state time
(5)	Programmed state time
(6)	Reset

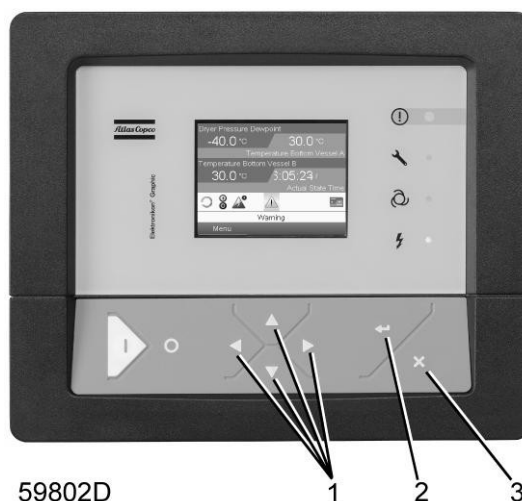
The screen shows a list of all counters with their actual readings. A number of counters keep track of the state of the dryer:

- Running hours:** counts the operation hours of the dryer.
- Loaded hours:** the same as running hours, but shows to which time interval the energy counters refer; this counter is also reset when the energy counters are reset.
- Actual state time:** shows how long the dryer's current state has been active.
- Programmed state time:** shows how long (at most) the current state should be active.
- Actual half cycle time:** shows how long the adsorbing vessel has been adsorbing (since last vessel shift).

- **Programmed half cycle time:** shows how long the half cycle should take; minimum value if PDP control is not active, maximum value if PDP is active.
- **Regeneration cycles vessel A:** integer that counts how many cycles vessel A has performed.
- **Regeneration cycles vessel B:** integer that counts how many cycles vessel B has performed.
- **Energy saving PDP sensor:** shows how much energy is saved (in %) by having the PDP sensor since the energy counters have been reset, so it equals StandbyTime/LoadedTime.
- **Module hours:** shows how long the Elektronikon controller has been active. This timer can not be reset, not even when downloading new Elektronikon software.

3.10 EVENT HISTORY MENU

CONTROL PANEL



59802D

(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, EVENT HISTORY



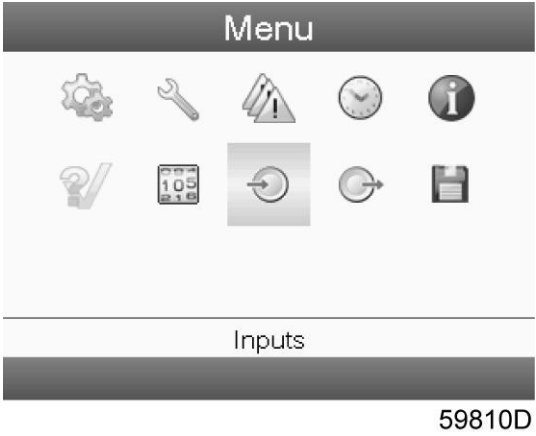
FUNCTION

To call up the last shut-down and last emergency stop data.

PROCEDURE

Starting from the Main screen (see [Main screen](#)),

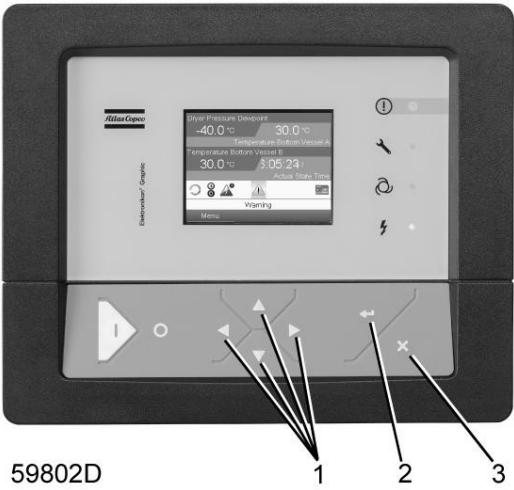
- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



- Using the Scroll keys, move the cursor to the Event History icon (see above, section Menu icon).
- The list of last shut-down and emergency stop cases is shown.
- Scroll through the items to select the desired shut-down or emergency stop event.
- Press the Enter key to find the date, time and other data reflecting the status of the dryer when that shut- down or emergency stop occurred.

3.11 SERVICE MENU

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, SERVICE



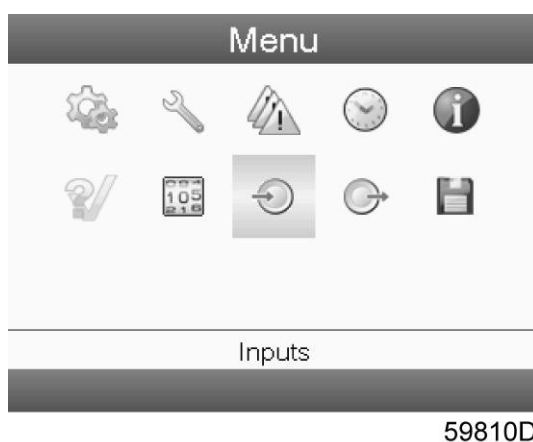
FUNCTION

- To reset the service plans which are carried out.
- To check when the next service plans are to be carried out.
- To find out which service plans were carried out in the past.
- To modify the programmed service intervals.

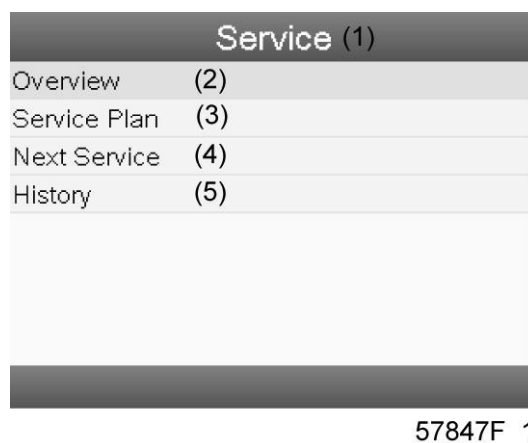
PROCEDURE

Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



- Using the Scroll keys, move the cursor to the Service icon (see above, section Menu icon).
- Press the Enter key. Following screen appears:



Text on figure

(1)	Service
(2)	Overview
(3)	Service plan
(4)	Next service
(5)	History

- Scroll through the items to select the desired item and press the Enter key to see the details as explained below.

OVERVIEW

Overview (1)				
4000		4000		A
4380		4337		A
8000		8000		B
8760		8717		B
40000		40000		C
43800		43757		C
(2) Running Hours		(3) Real Time Hours		

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Text on figure

(1)	Overview
(2)	Running Hours (green)
(3)	Real Time hours (blue)

Example for service level (A):

The figures at the left are the programmed service intervals. For Service interval A, the programmed number of running hours is 4000 hours (upper row, green) and the programmed number of real time hours is 4380 hours, which corresponds to six months (second row, blue). This means that the controller will launch a service warning when either 4000 running hours or 4380 real hours are reached, whichever comes first. Note that the real time hours counter keeps counting, also when the controller is not powered.

The figures within the bars are the number of hours to go till the next service intervention. In the example above, the dryer was just started up, which means it still has 4000 running hours or 4337 hours to go before the next Service intervention.

SERVICE PLANS

A number of service operations are grouped (called Level A, Level B, etc...). Each level stands for a number of service actions to be carried out at the time intervals programmed in the Elektronikon controller. When a service plan interval is reached, a message will appear on the screen.

After carrying out the service actions related to the indicated levels, the timers must be reset. From the Service menu above, select Service plan (3) and press Enter. Following screen appears:

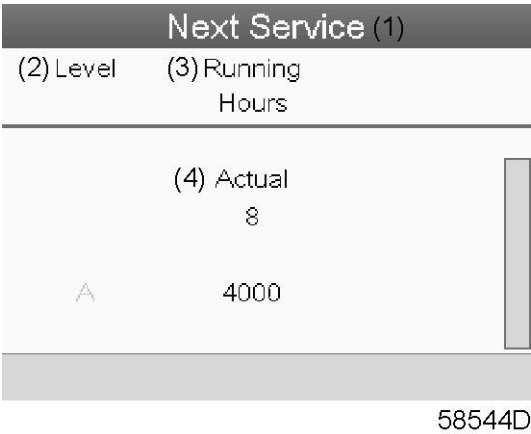
Service Plan (1)		
(2) Level	(3) Running Hours	(4) Real Time
A	4000	4380
B	8000	8760
C	40000	43800
D		
E		

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Text on figure

(1)	Service plan
(2)	Level
(3)	Running hours
(4)	Real time

NEXT SERVICE



Text on figure

(1)	Next service
(2)	Level
(3)	Running hours
(4)	Actual

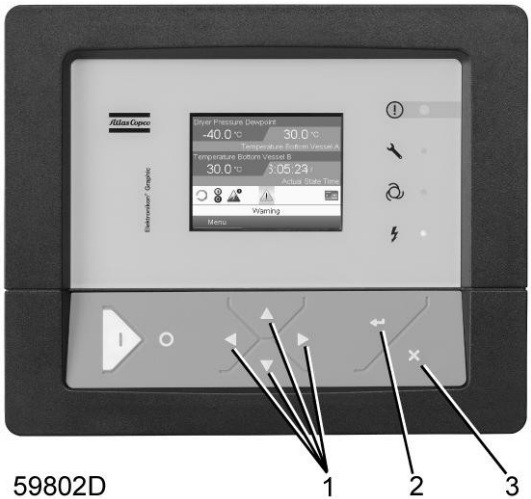
In the example above, the A Service level is programmed at 4000 running hours, of which 8 hours have passed.

HISTORY

The History screen shows a list of all service actions done in the past, sorted by date. The date at the top is the most recent service action. To see the details of a completed service action (e.g. Service level, Running hours or Real time hours), use the Scroll keys to select the desired action and press the Enter key.

3.12 PROTECTIONS MENU

CONTROL PANEL



MENU ICON, PROTECTIONS



FUNCTION

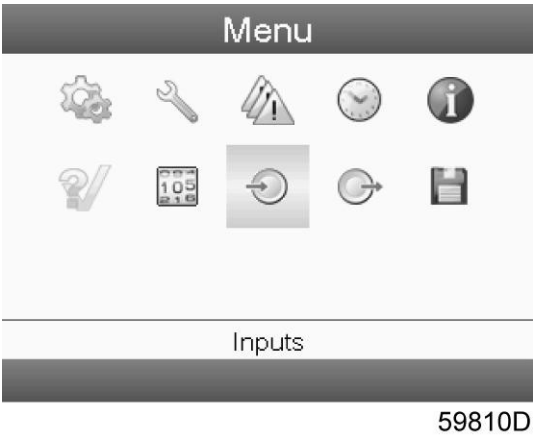
To call-up the protections.

	Before resetting a warning or shut down message, always remedy the problem. Frequently resetting these messages without remedying may damage the dryer.
---	--

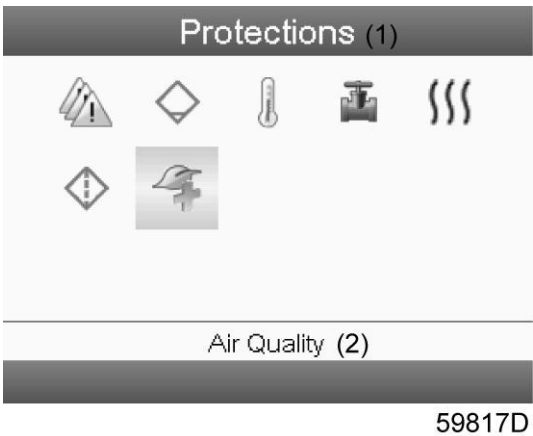
PROCEDURE

Starting from the Main screen (see [Main screen](#)):

- Move the cursor to the Menu action button and press the enter key (2). Following screen appears:



- Using the scroll keys (1), move the cursor to the protections icon (see above, section Menu icon).
- Press the enter key (2). Following screen appears:



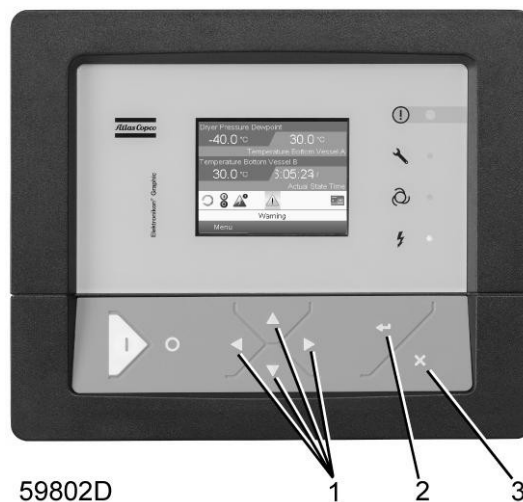
Check the *Icons used* section for the description of the icons. Text on figure

(1)	Protections
(2)	Air quality

- The screen shows a number of icons. Each icon indicates a menu item. By default, the Air Quality (2) icon is selected. The status bar shows the name of the menu that corresponds with the selected icon.
- Use the Scroll keys to select an icon.
- Press the Enter key (2) to open the menu.
- The screen shows a list of all protections with their corresponding icons and readings.
- If a protection is in warning or shutdown, the warning or shutdown icon blinks below the protection icon.

3.13 WEEK TIMER MENU

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, WEEK TIMER



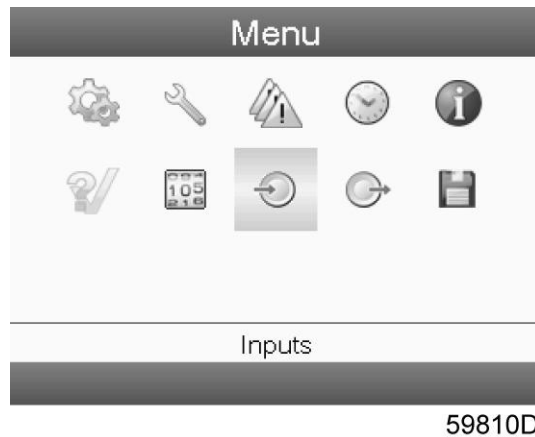
FUNCTION

- To program time-based start/stop commands for the dryer.
- To program time-based change-over commands for the net pressure band.
- Four different week schemes can be programmed.
- A week cycle can be programmed; a week cycle is a sequence of 10 weeks. For each week in the cycle, one of the four programmed week schemes can be chosen.

PROCEDURE

Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Use the Scroll buttons to select the Timer icon.



- Press the Enter key on the controller. Following screen appears:



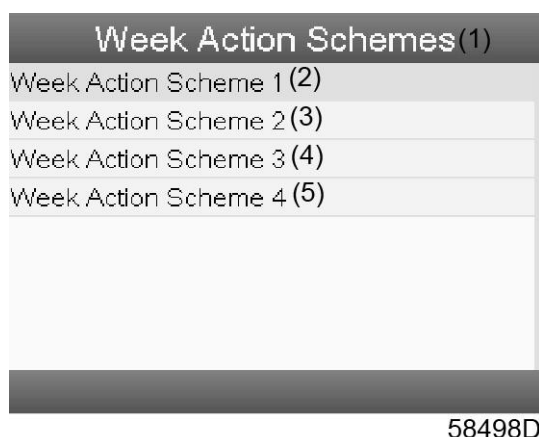
Text on figure

(1)	Week Timer
(2)	Week Action Schemes
(3)	Week Cycle
(4)	Status
(5)	Week 1
(6)	Remaining Running Time

The first item in this list is highlighted in red. Select the item requested and press the Enter key on the controller to modify.

PROGRAMMING WEEK SCHEMES

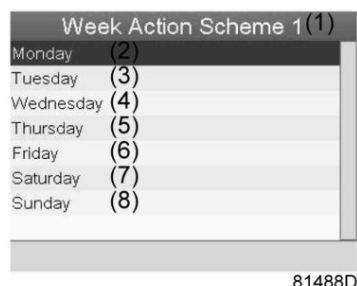
- Select Week Action Schemes and press Enter. A new window opens. The first item in the list is highlighted in red. Press the Enter key on the controller to modify Week Action Scheme 1.



Text on figure

(1)	Week Action Schemes
(2)	Week Action Scheme 1
(3)	Week Action Scheme 2
(4)	Week Action Scheme 3
(5)	Week Action Scheme 4

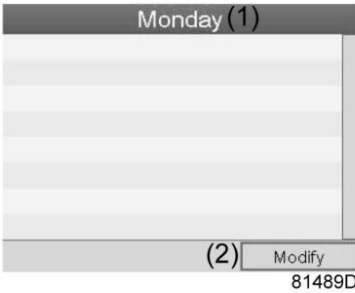
- A weekly list is shown. Monday is automatically selected and highlighted in red. Press the Enter key on the controller to set an action for this day.



Text on figure

(1)	Week Action Scheme 1
(2)	Monday
(3)	Tuesday
(4)	Wednesday
(5)	Thursday
(6)	Friday
(7)	Saturday
(8)	Sunday

- A new window opens. The Modify action button is selected. Press the Enter button on the controller to create an action.

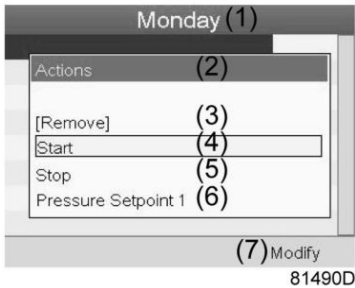


Text on figure

(1)	Monday
(2)	Modify

- A new pop-up window opens. Select an action from this list by using the Scroll keys on the controller.

When ready press the Enter key to confirm.



Text on figure

(1)	Monday
(2)	Actions
(3)	Remove
(4)	Start
(5)	Stop
(6)	Pressure Set point 1
(7)	Modify

- A new window opens. The action is now visible in the first day of the week.



Text on figure

(1)	Monday
(2)	Start

(3)	Save
(4)	Modify

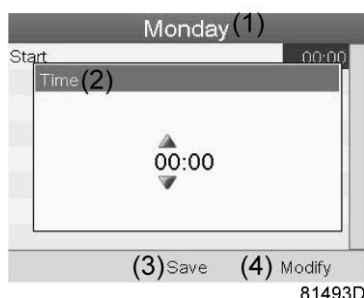
- To adjust the time, use the Scroll keys on the controller and press the Enter key to confirm.



Text on figure

(1)	Monday
(2)	Start
(3)	Save
(4)	Modify

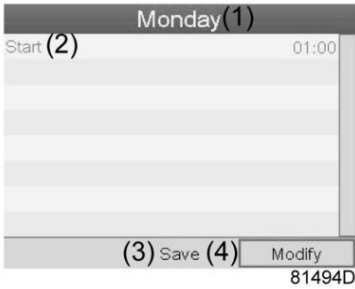
- A pop-up window opens. Use the ↑ or ↓ key of Scroll keys to modify the values of the hours. Use the ← or → Scroll keys to modify the minutes.



Text on figure

(1)	Monday
(2)	Time
(3)	Save
(4)	Modify

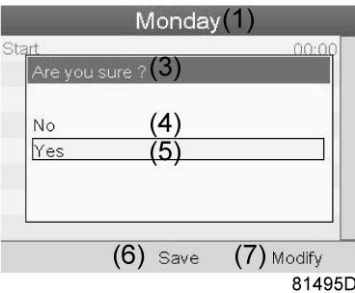
- Press the Escape key on the controller. The action button Modify (4) is selected. Use the Scroll keys to select the action Save.



Text on figure

(1)	Monday
(2)	Start
(3)	Save
(4)	Modify

- A new pop-up window opens. Use the Scroll keys on the controller to select the correct actions. Press the Enter key to confirm.

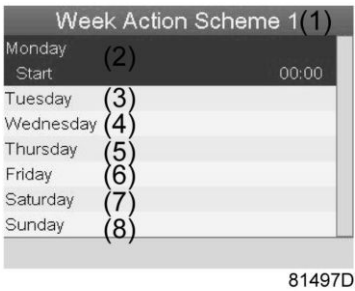


Text on figure

(1)	Monday
(3)	Are you sure?
(4)	No
(5)	Yes
(6)	Save
(7)	Modify

Press the Escape key to leave this window.

- The action is shown below the day the action is planned.



Text on figure

(1)	Week Action Scheme 1
(2)	Monday - Start
(3)	Tuesday
(4)	Wednesday
(5)	Thursday
(6)	Friday
(7)	Saturday
(8)	Sunday

Press the Escape key on the controller to leave this screen.

PROGRAMMING THE WEEK CYCLE

A week cycle is a sequence **of 10 weeks**. For each week in the cycle, one of the four programmed week schemes can be chosen.

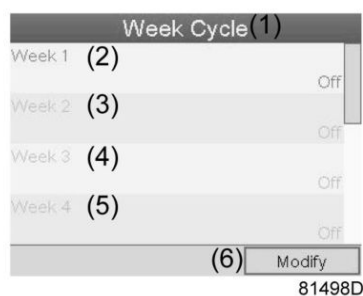
- Select Week Cycle from the main Week Timer menu list.



Text on figure

(1)	Week Timer
(2)	Week Action Schemes
(3)	Week Cycle
(4)	Status
(5)	Week Timer Inactive
(6)	Remaining Running Time

- A list of 10 weeks is shown.

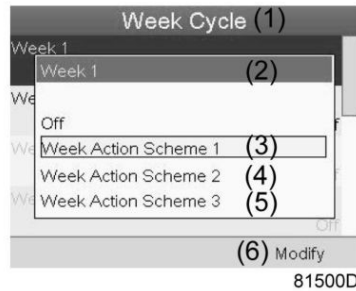


Text on figure

(1)	Week Cycle
(2)	Week 1
(3)	Week 2
(4)	Week 3
(5)	Week 4
(6)	Modify

Press the Enter key on the controller twice, to modify the first week.

- A new window opens. Select the action, example: Week Action Scheme 1



Text on figure

(1)	Week Cycle
(2)	Week 1
(3)	Week Action Scheme 1
(4)	Week Action Scheme 2
(5)	Week Action Scheme 3
(6)	Modify

- Check the status of the Week Timer

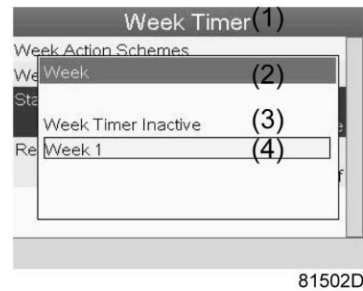
Use the Escape key on the controller to go back to the main Week Timer menu. Select the status of the Week Timer.



Text on figure

(1)	Week Timer
(2)	Week Action Schemes
(3)	Week Cycle
(4)	Status
(5)	Week Timer Inactive
(6)	Remaining Running Time

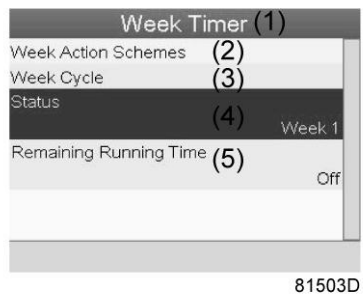
- A new window opens. Select Week 1 to set the Week Timer active.



Text on figure

(1)	Week Timer
(2)	Week
(3)	Week Timer Inactive
(4)	Week 1

- Press the Escape key on the controller to leave this window. The status shows that week 1 is active.



Text on figure

(1)	Week Timer
(2)	Week Action Schemes
(3)	Week Cycle
(4)	Status
(5)	Remaining Running Time

- Press the Escape key on the controller to go to the main Week Timer menu. Select Remaining Running Time from the list and press the Enter key on the controller to Modify.



Text on figure

(1)	Week Timer
(2)	Week Action Schemes
(3)	Week Cycle
(4)	Status
(5)	Remaining Running Time

- This timer is used when the week timer is set and for certain reasons the dryer must continue working. The remaining running time, for example 1 hour, can be set in this screen. This timer is prior to the Week Timer action.

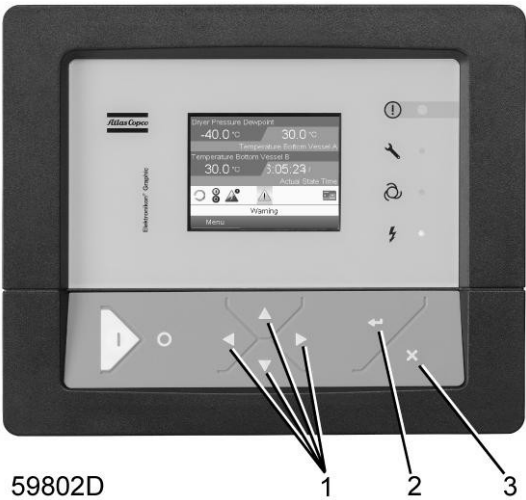


Text on figure

(1)	Week Timer
(2)	Week action schemes
(3)	Remaining Running Time

3.14 INFO MENU

CONTROL PANEL



(1)	Scroll keys
(2)	Enter key
(3)	Escape key

MENU ICON, INFO



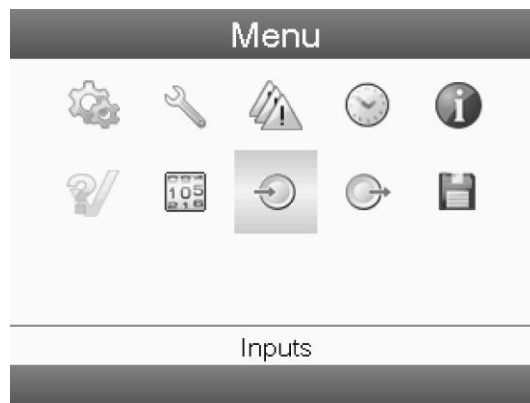
FUNCTION

To show the Atlas Copco internet address.

PROCEDURE

Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:

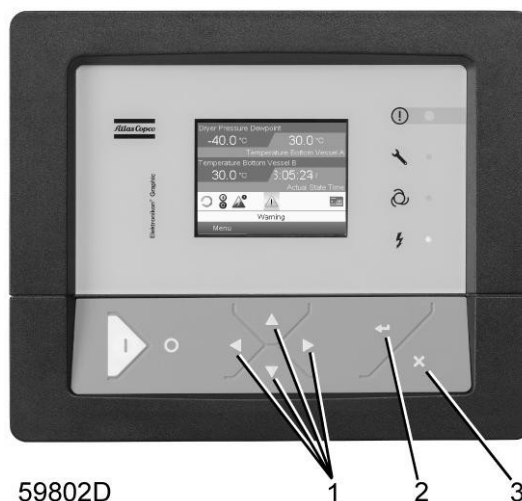


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- Using the Scroll keys, move the cursor to the Info icon (see above, section Menu icon).
- Press the Enter key. The Atlas Copco internet address appears on the screen.

3.15 MODIFYING SETTINGS

CONTROL PANEL



59802D

(1)	Enter key
(2)	Escape key
(3)	Scroll keys

MENU ICON, SETTINGS



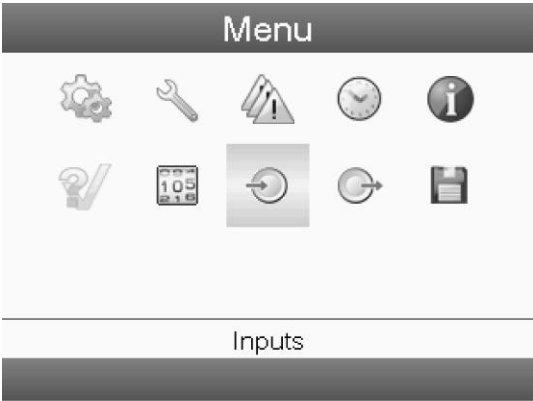
FUNCTION

To display and modify a number of settings (e.g. Time, Date, Date format, Language, units ...).

PROCEDURE

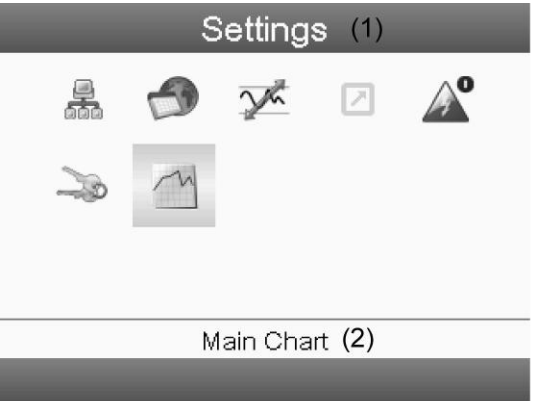
Starting from the Main screen (see [Main screen](#)),

- Move the cursor to the Menu action button and press the Enter key. Following screen appears:



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- Using the Scroll keys, move the cursor to the Settings icon (see above, section menu icon).
- Press the Enter key. Following screen appears:



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- The screen shows a number of icons:

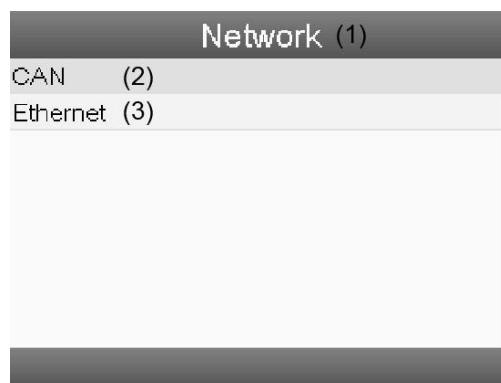
Icon	Function
------	----------

	Network settings
	General settings
	Regulation settings
	Automatic restart after voltage failure settings
	Access key
	Main chart

- Move the cursor to the icon of the function to be modified and press the Enter key.

MODIFYING NETWORK SETTINGS

- Select the network settings icon as described above and press the Enter button (2). Following screen appears:



58466D_1

Text on figure

(1)	Network
(2)	CAN
(3)	Ethernet

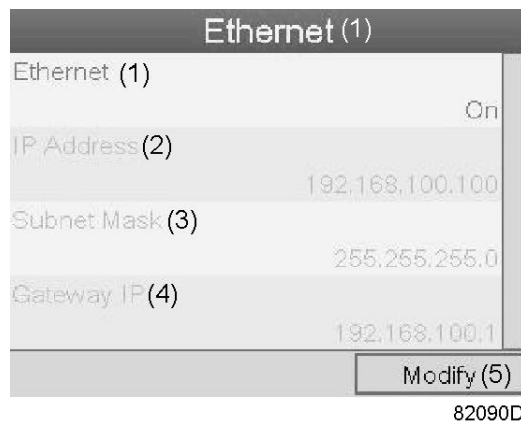
- A red selection bar is covering the first item (CAN). Use the ↓ key of the Scroll keys to select the setting to be modified and press the Enter key. Following screen appears:



Screen for CAN settings

Text on figure

(1)	CAN
(2)	CAN Address
(3)	PC Tools Channel
(4)	ES Channel
(5)	Modify



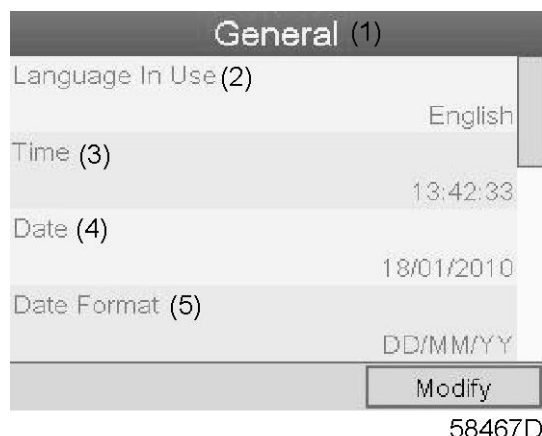
Screen for Ethernet settings

Text on figure

(1)	Ethernet
(2)	IP Address
(3)	Subnet mask
(4)	Gateway IP
(5)	Modify

- Press the Enter button; a red selection bar is covering the first item (CAN).
- Using the Scroll keys, move the cursor to the setting to be modified (e.g. Ethernet) and press the Enter button (2).
- A pop-up screen appears. Use the ↑ or ↓ key to select the required parameter and press the Enter key to confirm.

- Select the General settings icon as described above and press the Enter button (2). Following screen appears:



Text on figure

(1)	General
(2)	Language in use
(3)	Time
(4)	Date
(5)	Date format

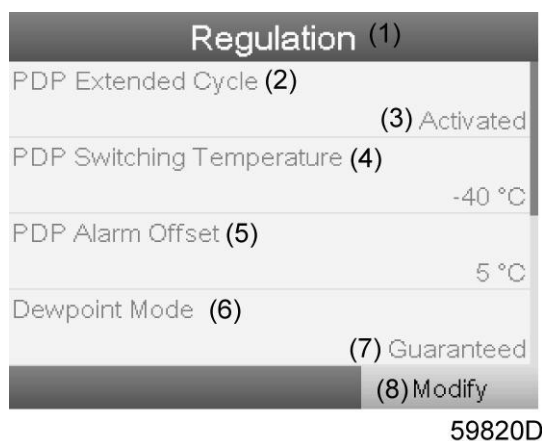
- The screen shows the first items of a list of all settings. Use the scroll button to see the other items in the list.
- Press the Enter button (2); a red selection bar is covering the first item (Language in use).

Use the ↓ key of the Scroll keys to select the setting to be modified and press the Enter key.

- A pop-up screen appears. Use the ↑ or ↓ key to select the required parameter and press the Enter key to confirm.

REGULATION SETTINGS

- Select the Regulation icon as described above and press the Enter button (2). Following screen appears:



Text on figure

(1)	Regulation
(2)	PDP extended cycle (PDP = Pressure Dew Point)
(3)	Activated
(4)	PDP switching temperature
(5)	PDP Alarm offset
(6)	Dew point mode
(7)	Guaranteed (or Suppressed, depending on its setting)
(8)	Modify

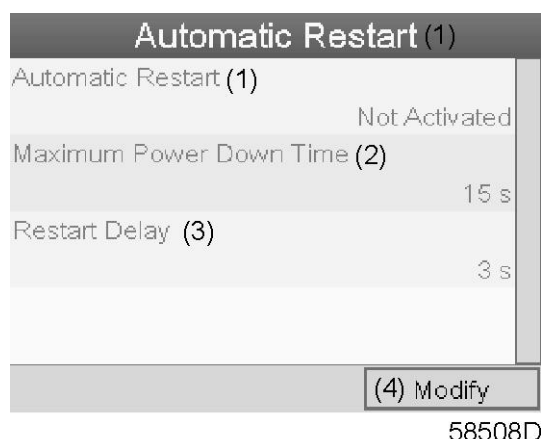
- The screen shows the list of all settings.
- Press the Enter button (2); a red selection bar is covering the first item (PDP extended cycle).
Use the ↓ key of the Scroll keys to select the setting to be modified and press the Enter key.
- A pop-up screen appears. Use the ↑ or ↓ key to select the required parameter and press the Enter key to confirm.

Regulation settings:

- **PDP extended cycle:** the cycle is lengthened by putting the vessel in waiting mode (standby) if PDP is low enough after regeneration and cooling.
- **PDP switching temperature:** the PDP should not increase to this setting before the vessels are switched when the dryer is in standby mode.
- **PDP alarm offset:** the PDP alarm is active when: $PDP > PDP_switching_temperature + PDP_alarm_offset$. In that way, the alarm threshold can be set independently of the switching threshold.
- **Dew point mode:** only available on Guaranteed type dryers (refer to [Introduction](#) section). If the dew point mode is set to Suppression, the dryer will regenerate without the *Regeneration without heaters* step.

AUTOMATIC RESTART

- Select the Automatic restart settings icon as described above and press the Enter button (2).
Following screen appears:



Text on figure

(1)	Automatic restart
(2)	Maximum power down time
(3)	Restart delay
(4)	Modify

- The screen shows the list of all settings.
- Press the Enter button (2); a red selection bar is covering the first item (Automatic restart).

Use the ↓ key of the Scroll keys to select the setting to be modified and press the Enter key.

- A pop-up screen appears. Use the ↑ or ↓ key to select the required parameter and press the Enter key to confirm.

ACCESS KEY

Different security levels are programmed in the regulator (e.g. user, service technician, etc). This menu item is used to change the security level. Scroll to the correct icon using the scroll key. Press the Enter button. Press the Enter button again to modify the security level. Press the Enter button again, a pop-up menu appears. Use the scroll keys to enter the password of the new security level. Press the Enter key to confirm the change.

MAIN CHART SETTINGS

	Chart ranges and bands can be modified. This can cause the current value to be out of range and can thus cause the absence of a visible curve in the graph.
---	---

The main chart settings menu allows to adjust the scale and curves on the graph. To modify these settings, proceed as follows:

- Select the Main chart settings icon as described above and push the Enter button (1). A screen similar to the one below appears:

Main Chart (1)	
Main Chart Signal (2)	Dp Air Filter
Chart Range (3)	
Minimum (4)	-0.172 bar
Maximum (5)	0.172 bar
Chart Band (6)	
Low (7)	(9) Off
High (8)	Off
Modify	

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Text on figure

(1)	Main chart
(2)	Main chart signal
(3)	Chart range
(4)	Minimum
(5)	Maximum
(6)	Chart band
(7)	Low
(8)	High
(9)	Off

The screen shows the main chart signal, the current chart range and band settings. To modify these settings, proceed as follows:

Push the Enter key, select chart range or chart band as desired and confirm.

MODIFYING THE CHART RANGE

Select the chart range as described above and proceed as follows:

- Push the Enter key; the minimum setting becomes highlighted. Push the Enter key to modify the minimum setting or use the arrow down key to modify the maximum setting.
- Push the Enter key to confirm the modification.

MODIFYING THE CHART BAND

Select the chart band as described above and proceed as follows:

- Push the Enter key; the low band setting becomes highlighted. Push the Enter key to modify the on/off setting or use the arrow down key to modify the low setting.
- Push the Enter key to confirm the modification.
- Proceed to modify the high band setting.

3.16 WEB SERVER

All Elektronikon controllers have a built-in web server that allows direct connection to the company network or to a dedicated PC via a local area network (LAN). This allows to consult certain data and settings via a PC instead of via the display of the controller.

GETTING STARTED

Make sure you are logged in as **administrator**.

- Use the internal network card from your computer or a USB to LAN adapter (see picture below).



USB to LAN adapter (for Windows XP)



59864F

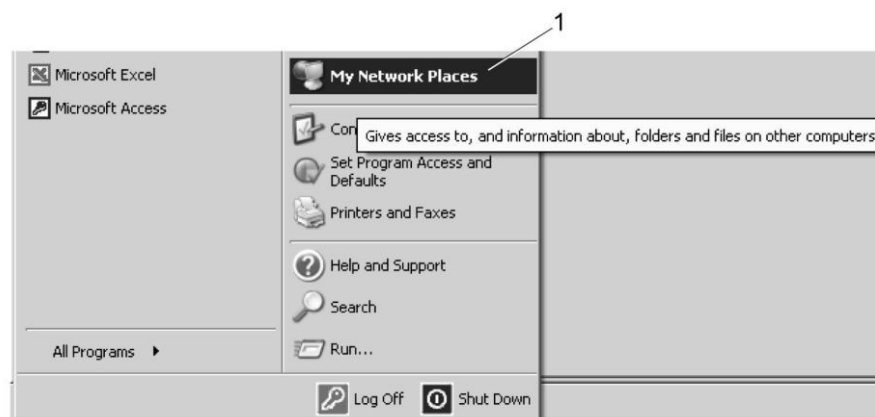
USB to LAN adapter (for Windows 7)

- Use a UTP cable (CAT 5e) to connect to the controller (see picture below).



CONFIGURATION OF THE NETWORK CARD

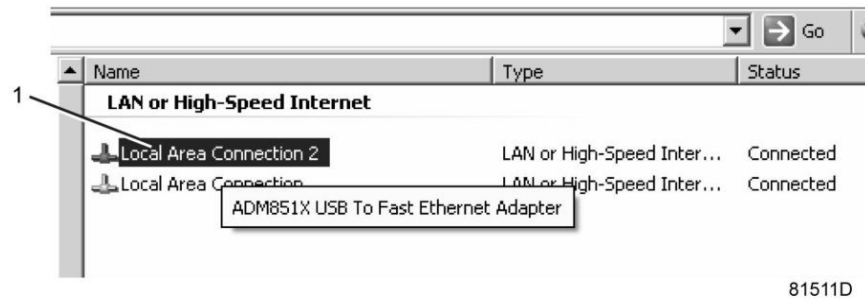
- Go to My Network places (1).



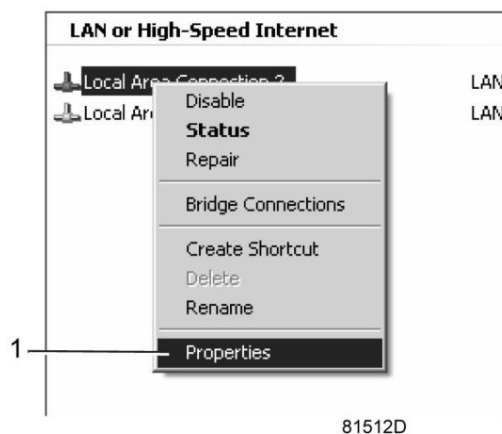
- Click on View Network connections (1).



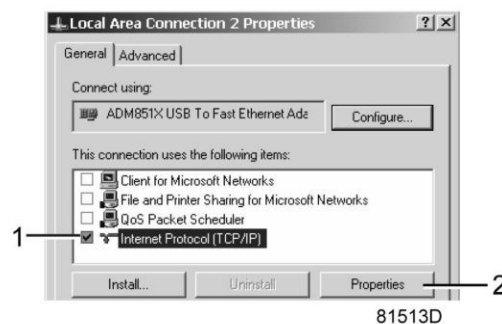
- Select the Local Area connection (1), which is connected to the controller.



- Click with the right button and select properties (1).



- Use the checkbox Internet Protocol (TCP/IP) (1) (see picture). To avoid conflicts, de-select other properties if they are selected. After selecting TCP/IP, click on the Properties button (2) to change the settings.



- Use the following settings:
- IP Address 192.168.100.200
- Subnetmask 255.255.255.0

Click OK and close network connections.

CONFIGURATION OF THE WEB SERVER

Configure the web interface

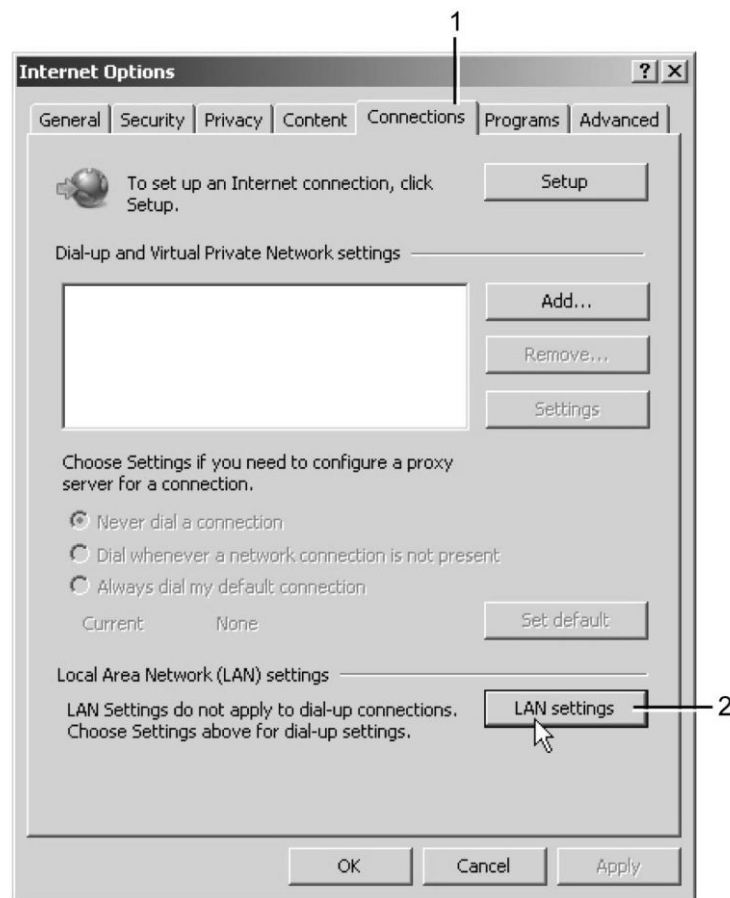


The internal web server is designed and tested for Microsoft® Internet Explorer 6, 7 and 8. Other web browsers like Opera and Firefox do not support this internal web server. When using Opera or Firefox, a redirect page opens. Click on the hyperlink to

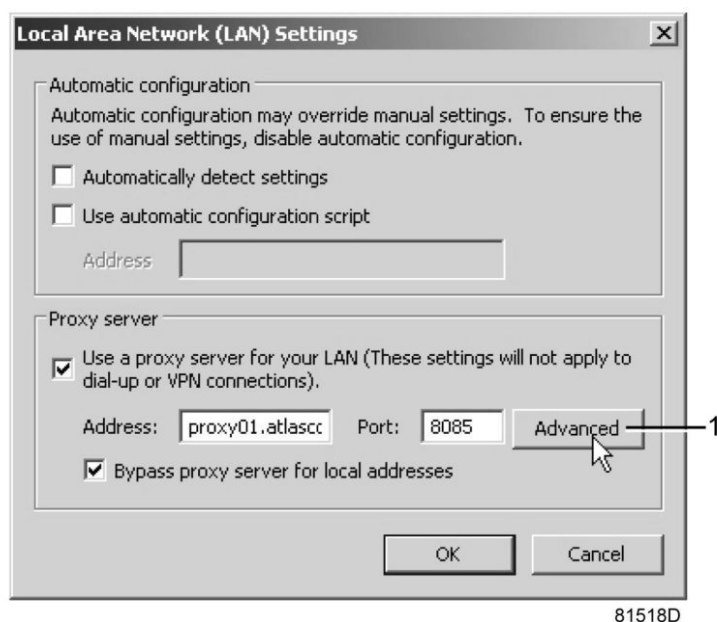
- When using Internet Explorer:
Open Internet Explorer and click on Tools - Internet options (2).



- Click on the Connections tab (1) and then click on the LAN settings button (2).

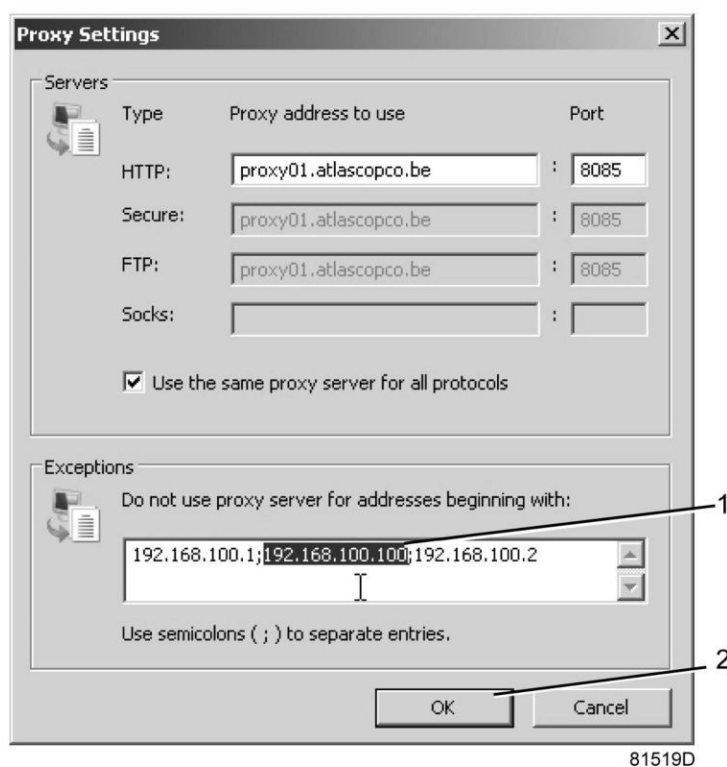


- In the Proxy server Group box, click on the Advanced button (1).



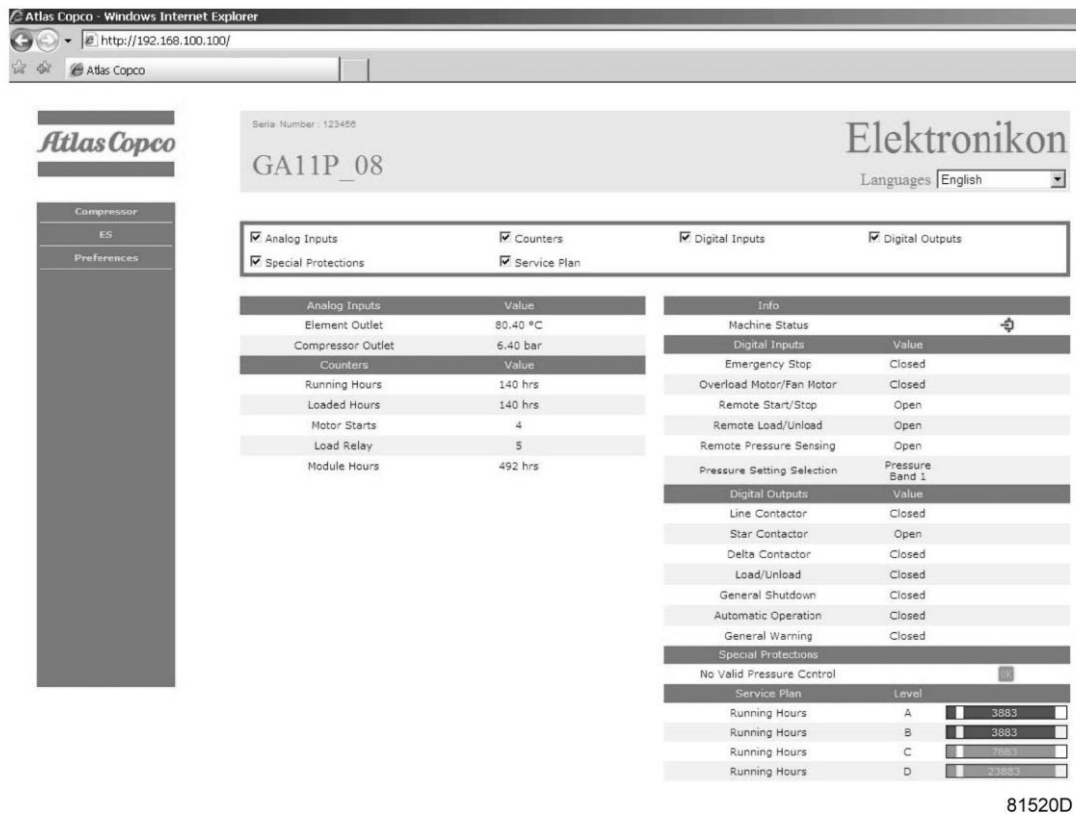
- In the Exceptions Group box, enter the IP address of your controller. Multiple IP addresses can be given but they must be separated with semicolons (;).

Example: Suppose that you already added two IP addresses (192.168.100.1 and 192.168.100.2). Now you add 192.168.100.100 and separate the 3 IP addresses by putting semicolons between them (1) (see picture). Click OK (2) to close the window.



VIEWING THE CONTROLLER DATA

- Open your browser and type the IP address of the controller you want to view in your browser (in this example <http://192.168.100.100/>). The interface opens:



Controller screen (typical).

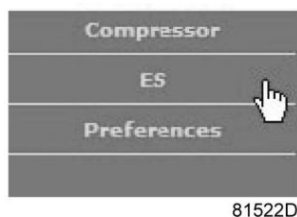
NAVIGATION AND OPTIONS

- The banner shows the dryer type and the language selector. In this example, three languages are installed on the controller.



- On the left side of the interface you can find the navigation menu (see picture below).
If a license for ESi is foreseen, the menu contains 3 buttons.
- dryer: shows all dryer settings.
- Es: shows the ESi status (if a license is provided).

- Preferences: allows to change temperature and pressure unit.



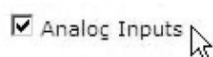
81522D

DRYER SETTINGS

All dryer settings can be hidden or shown. Put a mark for each setting. Only the machine status is fixed and can not be removed from the main screen.

Analog inputs

(The units of measure can be changed in the preference button from the navigation menu).

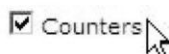


Analog Inputs	Value
Element Outlet	131.90 °F
Compressor Outlet	110.21 psi

81523D

Counters

Counters give an overview of all actual counters from controller and dryer.



Counters	Value
Running Hours	29 hrs
Loaded Hours	29 hrs
Motor Starts	3
Load Relay	4
Module Hours	549 hrs

81524D

Info status

Machine status is always shown on the web interface.



81525D

Digital inputs

Gives an overview of all Digital inputs and status.

☒ Digital Inputs

Digital Inputs	Value
Emergency Stop	Closed
Overload Motor/Fan Motor	Closed
Remote Start/Stop	Open
Remote Load/Unload	Open
Remote Pressure Sensing	Open
Pressure Setting Selection	Pressure Band 1

81526D

Digital outputs

Shows a list of all digital outputs and their status.

☒ Digital Outputs

Digital Outputs	Value
Line Contactor	Closed
Star Contactor	Open
Delta Contactor	Closed
Load/Unload	Closed
General Shutdown	Closed
Automatic Operation	Closed
General Warning	Closed

81527D

Special protections

Give an overview of all special protections of the dryer.

☒ Special Protections

Special Protections
No Valid Pressure Control

81528D

Service plan

Shows all levels of the service plan and status. This screen only shows the running hours. It is also possible to show the actual status of the service interval.

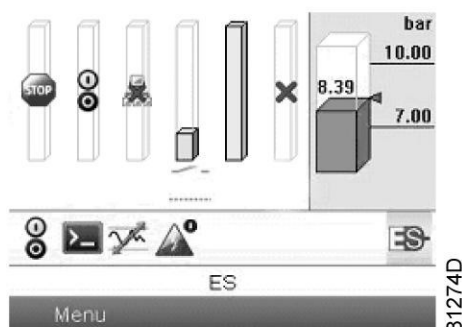
☒ Service Plan

Service Plan	Level	
Running Hours	A	3971
Running Hours	B	3971
Running Hours	C	7971
Running Hours	D	23971

81529D

ES screen controller

If an ESi license is provided, the button ES is shown in the navigation menu. At the left all dryers in the ES and at the right the ES status are shown.



A possible ESi screen

3.17 PROGRAMMABLE SETTINGS

DESCRIPTION

The regulation and safety devices are factory-adjusted to obtain optimum performance of the dryer. No adjustments are required.

4 INSTALLATION

4.1 DIMENSION DRAWINGS

DIMENSIONS

Dimension drawing of XD 1250 S

Dimension drawing of XD 1250 G

Dimension drawing of XD 1600 S

Dimension drawing of XD 1600 G

Dimension drawing of XD 2000 S

Dimension drawing of XD 2000 G

Dimension drawing of XD 2500 S

Dimension drawing of XD 2500 G

Dimension drawing of XD 3600 S

Dimension drawing of XD 3600 G

Dimension drawing of XD 4150 S

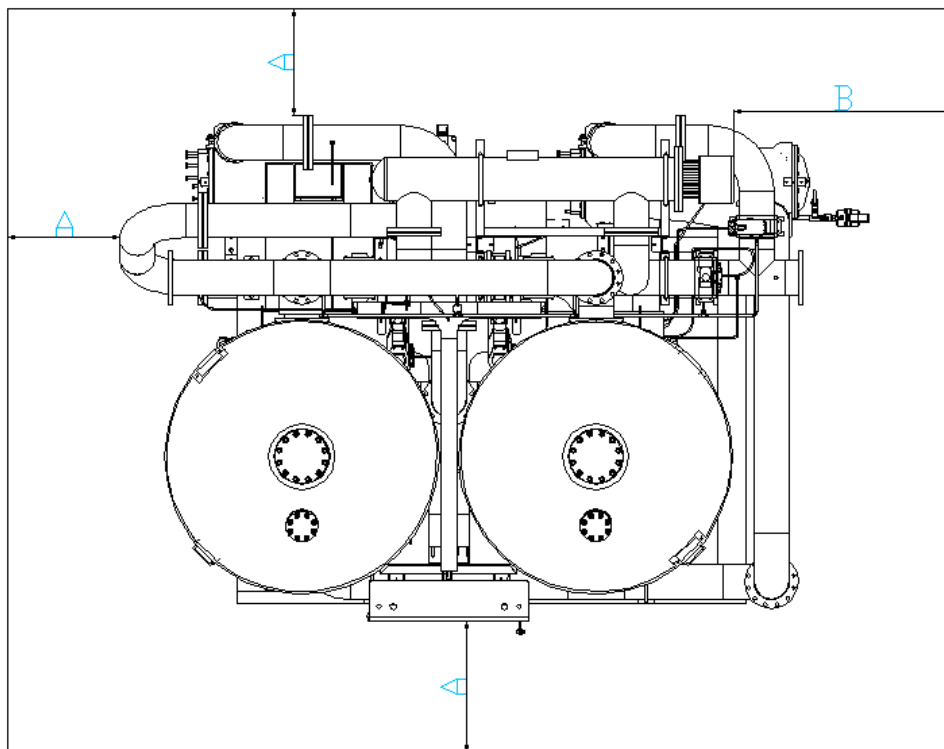
Dimension drawing of XD 4150 G

Dimension drawing of XD 5000 S

Dimension drawing of XD 5000 G

4.2 INSTALLATION PROPOSAL

FREE ROOM FOR SERVICING

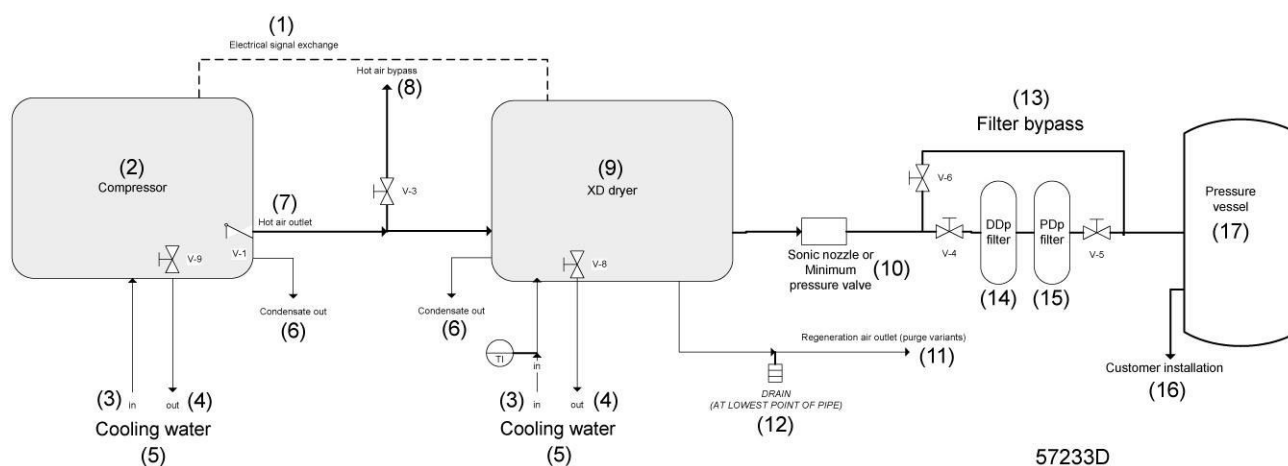


Free room around the dryer

Reference	Description
A	1000 millimeter
A	39.4 inch
B	1500 millimeter
B	59.1 inch

Refer to the *Dimension drawing* section for free room from the ceiling to be respected for servicing.

CONNECTIONS FROM COMPRESSOR TO AIR USER

*Installation proposal*

Reference	Description
V-1	Hot air outlet valve (compressor)
V-3	Hot air by-pass valve (compressor)
V-4	Filter inlet valve
V-5	Filter outlet valve
V-6	Filter by-pass valve
V-8	Cooling water outlet valve (dryer)
V-9	Cooling water outlet valve (compressor)
(1)	Electrical signal exchange
(2)	Compressor
(3)	In
(4)	Out
(5)	Cooling water
(6)	Condensate out
(7)	Hot air outlet
(8)	Hot air by-pass
(9)	XD dryer
(10)	Sonic nozzle or minimum pressure valve
(11)	Regeneration air outlet (Purge-cooled versions)
(12)	Drain (at lowest point of pipe)
(13)	Filter by-pass
(14)	DDp filter
(15)	PDp filter
(16)	Customer installation
(17)	Pressure vessel
(18)	Cold air outlet

4.3 TRANSPORT INSTRUCTIONS

INSTRUCTIONS

When transporting the dryer, the following point must be kept in mind:

- The bottom drain outlet of the cooler(s) must be opened and the drain plug removed to prevent water staying in the cooler(s) for a long period.

ATTENTION



If the dryer will not be used for a long period, contact your Atlas Copco Customer Centre for detailed instructions.

4.4 INSTALLATION INSTRUCTIONS

WARNING



Before switching on the main power supply, check the voltage requirements in the technical specifications or on the dryer data plate.

PROCEDURE

Item	Action
Dryer	- The unit should be installed on a level floor capable of taking the weight of the dryer. - Install the dryer under a roof, where the ambient air is as clean as possible and where the temperature of the air will never exceed the limits. Refer to the Reference conditions and limitations section. - Enough space must be provided around dryer and filters to allow maintenance,
Regeneration/ cooling air	- It is desirable to lead the regeneration/cooling outlet air via a pipe to another location; it is important to take into consideration the maximum allowable pressure drop. If the length of the additional pipe exceeds 5 m (16.405 ft), the diameter must be 1.5 times larger than the original regeneration pipe. For longer pipe installations, consult Atlas Copco. - If the pipe is led upwards, provide a drain hole at the lowest point to remove
Air	- Connect the compressed air lines to the marked inlet and outlet pipes of the dryer. Blow any welding waste and other impurities from the lines before connecting them. - The air inlet and outlet pipes should be installed stress free to the dryer unit. - Provide an air inlet valve and outlet valve to avoid inverse flow.
Drain pipes	- Each drain pipe must be individually connected to the waste pipe. - Lay out the condensate drain pipes via a funnel towards the waste pipe to allow visual inspection. The pipes must slope downwards. - If the condensate drain pipes have been fitted outside the compressor room
Cooling	- Connect the water cooling pipes to the marked inlet and outlet connections. - It is recommended to fit a manual shut-off valve and drain in the water inlet and
Hot surfaces	- Adequate protection must be provided to protect personnel from touching the hot surfaces of the dryer. - All inlet piping of the dryer will reach temperatures up to 150 °C (302 °F). Special precautions must be taken to prevent personnel from touching the hot surface.
Filters	After initial filling or replacement of the drying medium, the amount of dust released can increase. Therefore, it is recommended to replace the dust filter one week after replacing the desiccant and close the air inlet for the PDP sensor for the first 24 hours.

Item	Action
Electrical connection	• Check that the primary side connections of transformers (T1 and T2) correspond with the supply voltage. • Check if the overheating switches in the heater boxes on top of the vessels are set to 450 °C (842 °F). (G-type) • Refer to the Electric cable section, settings of circuit breakers and fuses section for the recommended fuses and cable size of the supply cables. • Check that the electrical installation corresponds to the local codes. The dryer

	• Always isolate the dryer inlet and outlet during maintenance. • When installing the piping, make sure all pipes are clean. • All pipes should be installed stress free to the dryer unit. If necessary, additional flexible connections and supports must be used.
---	--

TIGHTENING TORQUES

Dimension	Tightening torque (Nm)	Allowed deviation (+/- Nm)
M3	1	0.3
M4	2.4	0.6
M5	5	1.2
M6	8	2.1
M8	20	5
M10	41	10
M12	73	18
M14	115	29
M16	185	46
M18	238	60
M20	335	84
M22	462	116
M24	600	150
M27	858	214
M30	1200	300
M33	1581	395
M36	2000	500

4.5 ELECTRIC CABLE SIZE, SETTINGS OF CIRCUIT BREAKERS AND FUSES

IMPORTANT

	Local regulations remain applicable if they are stricter than the values proposed. The voltage drop must not exceed 5 % of the nominal voltage. It may be necessary to use cables with a larger size than those stated to comply with this requirement. For IEC approved dryers, the settings of the main fuses below are according to Directive 73/23/EEC (low-voltage directive) EN60204, 83/336/EC electromagnetic compatibility. The cable size is valid for cable PVC 70 °C (158 °F) at an ambient
---	--

CABLE SIZES AND FUSES

IEC, 380V, Zero Purge, 50 °C, G-type

Type	QF1 (A)	QF2 (A)	F1 (A)	F2 (A)	Max fuse (A)	Suggested cable (mm ²)
XD1250	50	50	--	--	125	4x35+25,
XD1600	60	60	--	--	150	4x50+35
XD2000	80	80	--	--	200	4x50+35
XD2500	80	80	--	--	200	4x50+35
XD3600	60	60	60	60	300	4x95+70
XD4150	80	80	80	80	400	4x120+95
XD5000	100	100	100	100	500	4x150+120

IEC, 220V, Zero Purge, 50 °C, S-type

Type	QF1 (A)	QF2 (A)	F1 (A)	F2 (A)	Max fuse (A)	Suggested cable (mm ²)
XD1250	--	--	--	--	10	3x6+6
XD1600	--	--	--	--	10	3x6+6
XD2000	--	--	--	--	10	3x6+6
XD2500	--	--	--	--	10	3x6+6
XD3600	--	--	--	--	10	3x6+6
XD4150	--	--	--	--	10	3x6+6
XD5000	--	--	--	--	10	3x6+6

4.6 COOLING WATER REQUIREMENTS

RECOMMENDATIONS

The cooling water quality must meet certain minimum requirements.

No general recommendation can encompass the effects of all combinations of the various compounds, solids and gases typically found in cooling water in interaction with different materials.

Only non-treated water must be used.

This recommendation is a general guide line for acceptable coolant quality.

TYPE OF SYSTEM

First, it is important to consider whether you are dealing with a closed system or an open system. In a closed system, the same cooling water circulates through a system without contact with air.

An open system is a pass-through system, or a circulating system with a cooling tower. In the latter case, the composition of the water that enters the cooler must be considered, and not the composition of the make-up water. Due to the evaporative effect in the cooling tower, much higher concentrations of ions can be obtained in the circulating water than in the make-up water.

RYZNAR STABILITY INDEX (RSI)

The Ryznar Stability Index (RSI) is a parameter for predicting whether water will tend to dissolve or precipitate calcium carbonate. The adhesion of scaling deposits and their effect are different on different materials, but the equilibrium of the water (scaling or corrosive) is only determined by its actual pH value and by the saturation pH value (pH_s).

The saturation pH value is determined by the relationship between the calcium hardness, the total alkalinity, the total solids concentration and the temperature.

The Ryznar Index is calculated as follows : $RSI = 2 \cdot pH_s - pH$

Symbol	Explanation
pH	Measured pH (at room temperature) of water sample
pH_s	pH at saturation

The pH_s is calculated by using :

$$pH_s = (9.3 + A + B) - (C + D)$$

Symbol	Explanation
A	Depends on the total solids concentration (mg/l)
C	Depends on the calcium hardness (ppm $CaCO_3$)

D	Depends on the HCO_3^- concentration or M-alkalinity (mval/l)
---	--

The values of A, B, C and D can be found in the following table.

Total dissolved solids (mg/l)	A	Temperature (°C)	B	Ca-hardness (ppm CaCO_3)	C	M-alkalinity (mval/l)	D
50 - 300	0.	0 - 1	2.	10 - 11	0.	0.20 - 0.22	1.
400-1000	0.	2 - 6	2.	12 - 13	0.	0.24 - 0.26	1.
		7 - 9	2.	14 - 17	0.	0.28 - 0.34	1.
		10 - 13	2.	18 - 22	0.	0.36 - 0.44	1.
		14 - 17	2.	23 - 27	1.	0.46 - 0.54	1.
		18 - 21	2.	28 - 34	1.	0.56 - 0.70	1.
		22 - 27	2.	35 - 43	1.	0.72 - 0.88	1.
		28 - 31	1.	44 - 55	1.	0.90 - 1.10	1.
		32 - 37	1.	56 - 69	1.	1.12 - 1.38	1.
		38 - 44	1.	70 - 87	1.	1.40 - 1.76	1.
		45 - 50	1.	88 - 110	1.	1.78 - 2.20	2.
		51 - 56	1.	111 - 138	1.	2.22 - 2.78	2.
		57 - 63	1.	138 - 174	1.	2.80 - 3.54	2.
		64 - 71	1.	175 - 220	1.	3.54 - 4.40	2.
		72 - 80	1.	230 - 270	2.	4.6 - 5.4	2.
				280 - 340	2.	5.6 - 7.0	2.
				350 - 430	2.	7.2 - 8.8	2.
				440 - 550	2.	9.0 - 11.0	2.
				560 - 690	2.	11.2 - 13.8	2.
				700 - 870	2.	14.0 - 17.6	2.
				880 - 1000	2.	17.8 - 20.0	3.

Interpretation of the value obtained

RSI	Water condition	Action
$\text{RSI} < 3.9$	Very high scale formation	Water cannot be used.
$4.0 < \text{RSI} < 5.5$	High boiler scale formation	Regular inspection and descaling operation necessary.
$5.6 < \text{RSI} < 6.2$	Slight boiler scale formation	Water treatment not necessary. Occasional inspection recommended.
$6.3 < \text{RSI} < 6.8$	Neutral water	Water treatment not necessary. Occasional inspection recommended.
$6.9 < \text{RSI} < 7.5$	Slight corrosion at higher temperature	Water treatment not necessary. Occasional inspection recommended.
$7.6 < \text{RSI} < 9.0$	Strong corrosion	Regular inspection necessary, use of corrosion inhibitor recommended.

RSI	Water condition	Action
$9.1 < \text{RSI} < 11$	Very strong corrosion	Regular inspection necessary, use of corrosion inhibitor required.
$\text{RSI} > 11$	Very strong corrosion in complete water system	Water should not be used.

The table indicates that distilled or demineralized water should never be used, as their RSI is > 11 .

The RSI only indicates the equilibrium of scaling - descaling. A cooling water showing good RSI conditions can still be unsuitable due to other factors.

From the table above, the RSI index should be between 5.6 and 7.5; otherwise, contact a specialist.

PH

TOTAL DISSOLVED SOLIDS (TDS)

This is the sum of all the ions in the water. It can be derived from the dry residue after evaporation (but not including suspended solids), or it can be estimated from the electrical conductivity.

In a closed system, the following limits apply : $\text{TDS} < 3000 \text{ mg/l}$ ($< 3800 \text{ microS/cm}$)

CHLORIDES (Cl^-)

Chloride ions will create pitting corrosion on stainless steel. Their concentration should be limited: Closed cooling system : chlorides $< 500 \text{ ppm}$

Open cooling system : chlorides $< 150 \text{ ppm}$

However, if the water is scaling, lower limits should be used. (See The Ryznar stability index (RSI)).

FREE CHLORINE (Cl_2)

A level of 0.5 ppm should not be exceeded for long periods.

For shock treatments, a maximum limit of 2 ppm for maximum 30 minutes/day applies.

SULPHATES (SO_4^{2-})

Closed cooling system : sulphates $< 400 \text{ ppm}$ Open cooling system : sulphates $< 150 \text{ ppm}$

CARBONATE HARDNESS

Closed cooling system : $50\text{-}1000 \text{ ppm CaCO}_3$ Open cooling system : $50\text{-}500 \text{ ppm CaCO}_3$

$\text{HCO}_3^- / \text{SO}_4^{2-}$ should be > 1

AMMONIA

< 0.5 ppm

COPPER

< 1 ppm

IRON AND MANGANESE

< 1 ppm

ORGANICS

No algae No oil

SUSPENDED SOLIDS

Non-soluble particles, size < 1 mm.

< 10 ppm

4.7 PICTOGRAPHS

EXPLANATION OF PICTOGRAPHS



1



3



2



4



7

Nm	M2,5	M3	M3,5	M4	M5	M6	M8	M10	M12
-	1.2	1.2	1.2	2.7	7	12	20	30	
0.5	0.5	0.8	1.2	-	3	6	10	15.5	

1078 9006 09

OK OK

8



9



10



5



6



11



12



13

57234P

Reference	Name
1	Cooling water outlet
2	Cooling water inlet
3	Manual drain
4	Automatic drain
5	Data plate
6	Disconnect from electrical supply before opening (Only for CSA/UL approved units)
7	Switch off the voltage and depressurise the dryer before maintenance or repair

Reference	Name
8	Tightening torques
9	Air outlet
10	Air inlet
11	Warning, heated surface
12	Warning, voltage
13	Dangerous blow-off, hot air

5 OPERATING INSTRUCTIONS

5.1 WARNINGS

SAFETY PRECAUTIONS

The operator must apply all relevant [Safety precautions](#), including those mentioned in this manual.

SAFETY VALVES

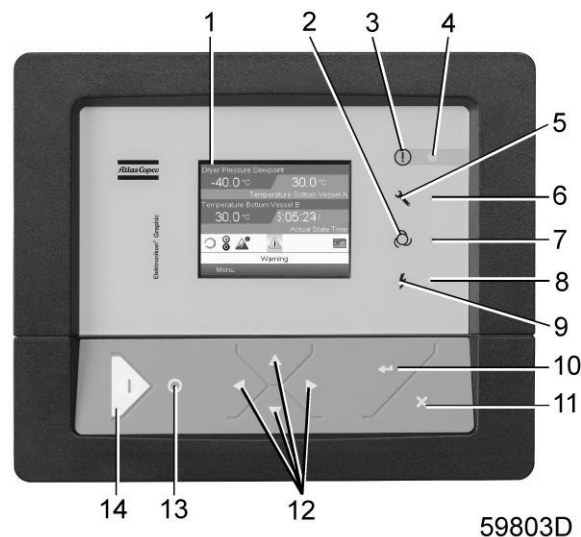
Safety valves on the inlet of the dryer are fitted as standard. These safety valves are sized for the maximum operating pressure and the nominal flow of the dryer.



The operator should ensure that the flow to the dryer can never exceed this nominal value. If not, the safety valves may not be able to blow off enough flow to properly

5.2 INITIAL START

PROCEDURE



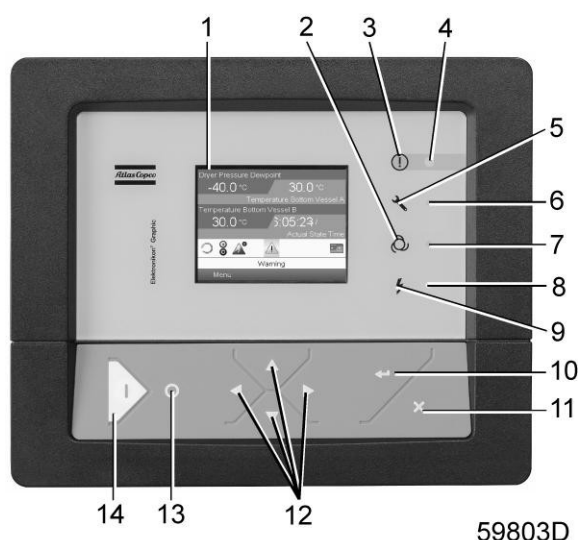
The dew point of the air leaving the dryer will be higher than normal after starting.

Step	Action
1	Check if the water plugs of the cooler(s) are installed.
2	Open the water inlet and outlet valves (customer's installation).
3	Close the air supply to the PDP sensor. (Optional for S-type)
4	- Switch on the voltage (customer's installation). - Switch on the voltage to the dryer by means of main switch (S1). (Except for dew

Step	Action
5	Check for sensor errors.
6	Press start button (14) on the control panel.
7	Start the compressor.
8	If no sonic nozzle or minimum pressure valve is installed. Slowly pressurize the dryer by closing the outlet valve or pressurize from the outlet side of the dryer.
9	If the air net is to remain dry, run the dryer approximately 7 hours (complete cycle for both towers) with the outlet air blown off to make sure that the desiccant is properly
10	Open the valve to the PDP sensor after 24 hours. (Optional for S-type)

5.3 STARTING

CONTROL PANEL



	Before starting, the dryer has its start conditions checked in the dryer starting routine of the Elektronikon. If these conditions are not fulfilled, the dryer will not start, the alarm LED (4) will light up after the permissive start time (180 seconds) and the cause will be indicated by an
--	---

PROCEDURE

Step	Action
1	Check if all fuses in the cubicle are turned on.
2	Switch on the voltage (customer's installation). The control panel shows that the dryer is stopped.
3	Switch on the voltage to the dryer by means of the main switch.
4	Check that the pilot air valve is open.

5	Check if the working pressure of the pneumatic filter/regulator is around 6 bar (87.02 psi).
---	--

Step	Action
6	Check if the cooling water supply is opened.
7	While the dryer outlet valve (customer's installation) is closed, open the air inlet valve (customer's installation) slowly and wait until the dryer has pressurized.
8	Check for leakage.
9	Start the dryer by pressing start button (14) on the control panel. If necessary, reset any alarm on the control panel.
10	Open the dryer air outlet valve slowly (customer's installation).
11	Close the by-pass valve (if provided).

DEWPOINT METER

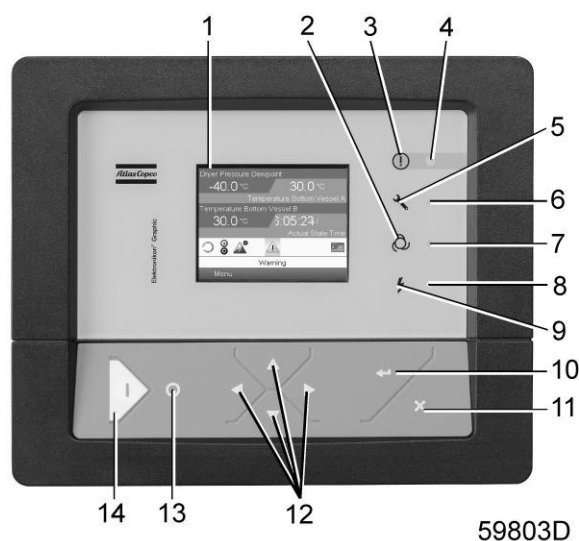
- It is recommended to run the dryer for 1 day before opening the valve to the dewpoint meter.
- The dryer must run in a fixed cycle for 1 day before switching over to -PDP control- (dewpoint meter control).

NOTE

	When the compressor runs unloaded, the dryer cycle will be halted temporarily, until the compressor loads again.
	A sonic nozzle can be provided to prevent starting problems (optional).

5.4 STOPPING

CONTROL PANEL



PROCEDURE

Step	Action
1	Press stop button (13). The dryer stops. Voltage on LED (8) remains lit. Leave the voltage on if the dryer has to remain on stand-by.
2	Close the dryer inlet and outlet valves (customer's installation).
3	Close the water inlet and outlet valves of the cooling water circuit (customer's installation). If the dryer is installed in a room where freezing temperatures are
4	For a long term stop: Remove the drain plugs in order to empty the cooler(s).

ATTENTION



Make sure that the outlet valve of the dryer is closed or the air net behind the dryer depressurizes slowly to prevent high air speeds and thus desiccant floating. A sonic

5.5 REMOTE START/STOP

ELECTRICAL CABINET



S3

59832F

The XD dryer can be started and stopped from a remote location by means of a digital switch S2 (customer's installation).

PROCEDURE

Step	Action
1	Stop the dryer. Refer to the <i>Stopping</i> section.
2	Switch off the voltage (customer's installation).
3	Connect a digital switch S2 (customer's installation) in between clamps 50 and 52 of rail 1X8 inside the cubicle. The switch should be of the Normally Open (NO) type.
4	Switch on the voltage (customer's installation)
5	Close the cubicle door.
6	Change the setting from <i>Local Control</i> to <i>Remote Control</i> . Refer to the <i>Control mode selection</i> section. Note that activating the <i>Remote Control</i> feature will disable the
7	To start the dryer push I on the Elektronikon regulator.

5.6 EMERGENCY STOP

ELECTRICAL CABINET



S3

59832F

In case of an emergency stop (S3):

The dryer will be shut down and the dryer cycle will pause.

To clear the shutdown:

1. Unlock the emergency stop button (S3).
2. Reset the shutdown alarm on the control panel.

WARNING

Do not use the emergency stop button to stop the unit in normal operation.

IMPORTANT

Note that the 24V AC control circuit of the Elektronikon regulator remains powered and the control panel will indicate an emergency stop shutdown.

5.7 VOLTAGE FAILURE

DESCRIPTION

In case of a voltage failure, the dryer will automatically restart if the supply voltage to the module is restored within a programmed time period (set ex-factory at 180 seconds).

Via the dryer starting routine the program will jump to the cycle step in which the voltage failure occurred. Refer to the *Dryer starting routine* section.

5.8 TAKING OUT OF OPERATION

PROCEDURE

At the end of the service life of the dryer, proceed as follows:

Step	Action
1	Close the air outlet valve and stop the dryer.
2	Press the emergency stop button.
3	Switch off the voltage and disconnect the dryer from the mains.
4	Isolate the dryer from the compressor and the air net.
5	Depressurise the dryer.
6	Disconnect the dryer air inlet and outlet pipe respectively from the compressor and the
7	Switch off and drain the cooling water and condensate circuits.
8	Disconnect the dryer condensate piping from the condensate drain net.
9	Disconnect the cooling water piping from the dryer.

6 MAINTENANCE

6.1 MAINTENANCE WARNINGS

WARNINGS



Before starting any maintenance or repairs, stop the dryer.
 Press the emergency stop button.
 Depressurise the air system.
 Always isolate the dryer inlet and outlet during maintenance.
 Open the isolating switch (customer's installation) to switch off the voltage to the dryer.
 Close the inlet and outlet valves of the cooling water supply.

6.2 PREVENTIVE MAINTENANCE SCHEDULE

WARRANTY AND PRODUCT LIABILITY

Use only Atlas Copco authorised genuine parts. Any damage or malfunction caused by the use of unauthorised parts is not covered by Warranty or Product Liability.

SERVICE PLAN

A number of service operations are grouped (called level I, level A, level B, ...). Each level stands for a number of service actions to be carried out at the time intervals programmed in the Elektronikon regulator.

When a level is reached, a message will appear on the screen. After carrying out all service actions, reset the interval timers using the "Reset" key (See section [Service menu](#)). If in any doubt, consult your Atlas Copco Customer Centre.

REGULAR SERVICE OPERATIONS

To ensure safe operation and long service life carry out the following operations at the interval (period or running hours) which comes first. The longer interval checks must also include the shorter interval checks.

The local Atlas Copco Customer Centre may overrule the maintenance schedule depending on the environmental and working conditions of the compressor.

When servicing, replace all removed gaskets, O-rings and washers.

Period	Service plan	Running hours	Operation
Daily	--	--	Check that the compressed air inlet temperature does not drop below the minimum value (see section

“	--	--	Check that dryer discharge pressure does not drop below minimum allowed pressure.
---	----	----	---

Period	Service plan	Running hours	Operation
“	--	--	Check for alarms on the display.
“	--	--	Check dewpoint.
“	--	--	Check the working of the EWD's.
Weekly	--	--	Check filter at outlet of dryer. (If optional filter is installed)
“	--	--	Check for water or air leaks.
“	--	--	Check for abnormal noise.
3-Monthly	I	2000	Check all temperatures and pressures.
“	“	“	Check filter of the pilot air circuit and clean/replace if necessary.
“	“	“	Check condensate drains, clean if necessary.
6-Monthly	A	4000	Check filter element of the PDP sensor block housing, replace if necessary.
“	“	“	Check/clean silencers, replace if necessary. (Except for zero purge cooled versions).
Yearly	B	8000	Replace the filter of the pilot air circuit.
“	“	“	Recalibrate/replace PDP sensor. (Optional)
Every 3	C	24000	Inspect cooler(s), clean if necessary.
Every 5	D	40000	Replace desiccant (if needed).
When displayed		--	Carry out the service actions in accordance with the displayed service plans.

NOTES

	Any leak should be attended to immediately.
	The interval depends on the working conditions of dryer and compressor. Consult your Atlas Copco Customer Centre for more information.

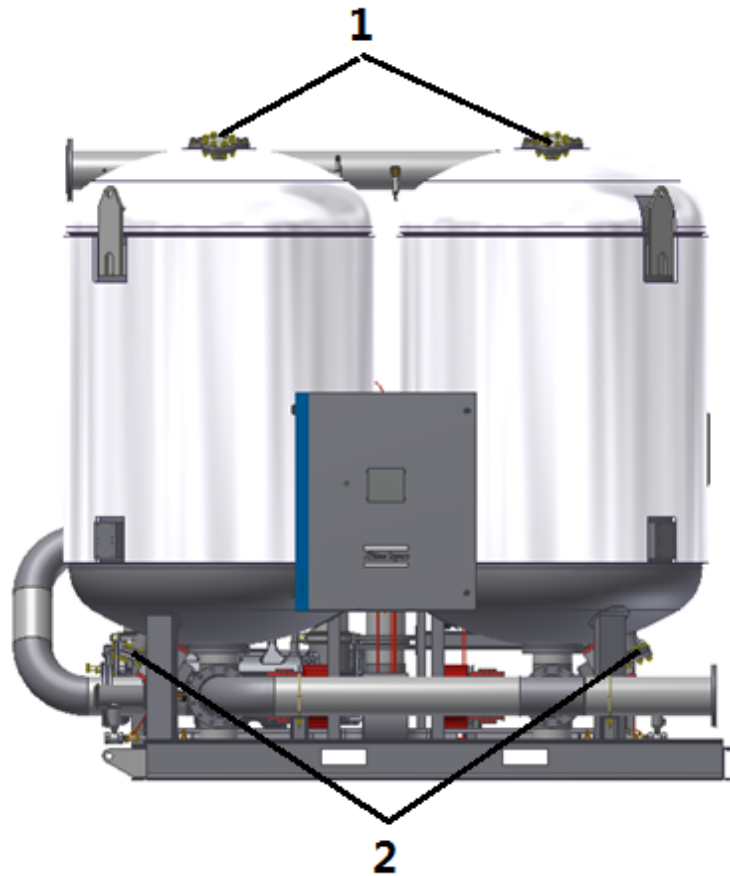
SERVICE AGREEMENTS

Atlas Copco Customer Centres have a range of service agreements to suit your needs:

- An Inspection Plan.
- A Preventive Maintenance Plan.
- A Total Responsibility Plan.

Contact your Customer Centre to arrange a tailor-made service agreement. This will ensure optimum operational efficiency, minimize downtime and reduce the total life cycle costs.

6.3 REPLACING DESICCANT



Removal and filling plugs

Reference	Description
1	Desiccant filling plugs (carrying the safety valves)
2	Desiccant removal plugs

EMPTYING

	Safety glasses, gloves and a dust mask must be worn during the emptying procedure as dust may be produced.
	When replacing desiccant, make sure that the desiccant is disposed of according to the local regulations.

Procedure

1. Let the dryer program run to the end of the cooling step or the standby / tower shifting step.
2. Stop the dryer.
3. Shut off the power supply.
4. Isolate the dryer from the air net by closing the air inlet and outlet valves.

5. Depressurize the vessels slowly.
6. Install a suitable collector underneath the desiccant removal plugs (2) for emptying; note that the desiccant may be very hot (150 °C (302 °F)).
7. Remove the desiccant removal plugs (2) and let the desiccant beads flow into the collector.

	A vacuum cleaner may be used to speed up the process and to control dust emission.
---	--

FILLING

	Safety glasses, gloves and a dust mask must be worn during the filling procedure as dust may be produced.
	Do NOT overfill the dryer.
	When replacing desiccant, make sure that the desiccant is disposed of according to the local regulations. When replacing the desiccant, never use an ejector or similar equipment as this might shatter the beads. The beads must be gravity-fed into the tower. After replacing the drying medium, the amount of dust released can increase.

Procedure

1. Once the vessels are empty, refit the desiccant removal plugs (2).
2. Open the desiccant filling plugs (1).
3. Fit a large funnel to the filling hole of one of the drying vessels; the size of the narrow part should be 35-40 mm (1.4 - 1.6 in).
4. Gently pour the correct amount of desiccant (refer to the *Air dryer data* section) into the vessel. The desiccant level must stay below the level of the top strainer.
Use a vacuum cleaner to control dust escaping from the filling hole.
5. Repeat steps 3 and 4 for the other tower; the amounts of desiccant must be equal in both vessels.
6. Refit the desiccant filling plugs (1). The dryer is now ready to be started. Refer to the *Starting* section.

NOTE

	To ensure PDP sensor functionality it is recommended to: • Run the dryer for 24 hours before opening the valve to the PDP sensor. • Run the dryer during an entire cycle (24 hours) before switching to PDP control
---	--

DESICCANT QUANTITIES

Depending on the dryer type, a quantity of water resistant Silica Gel beads is prescribed

Dryer size	Quantity (kg)	Quantity (lb)
XD1250	2x1000	2x2205
XD1600	2x1350	2x2976
XD2000	2x1700	2x3748
XD2500	2x2050	2x4519
XD3600	2x2750	2x6063
XD4150	2x3400	2x7496
XD5000	2x4100	2x9039

Ordering numbers and quantities of water resistant Silica Gel desiccant refill bags:

Ordering number	Quantity (kg)	Quantity (lb)
1624 6028 49	25	55
1624 6028 51	25	55

6.4 ELECTRONIC WATER DRAIN (EWD)

MAINTENANCE ACTIVITY

The set of wearing parts (service kit) must be changed every 8000 hours or yearly, whichever comes first.

SERVICE KITS (SET OF WEARING PARTS)

Service kit (set of wearing parts)	Ordering number
EWD DRAIN VALVE 550mL	1639 6141 35
EWD DRAIN VALVE 1300mL	1639 6141 39

6.5 SERVICE KITS

SERVICE KITS

Atlas Copco Customer Centres will be glad to provide you with a wide range of service kits. Service kits comprise all parts needed for servicing components and offer the benefits of genuine Atlas Copco parts while keeping the maintenance budget low. All service kits are indicated in the relevant Parts Lists.

7 OPTIONAL EQUIPMENT

7.1 PRECAUTIONS FOR OPTIONAL EQUIPMENT

WARNING



All responsibility for any damage or injury resulting from neglecting these precautions, or non-observance of the normal caution and care required for installation, operation, maintenance and repair, even if not expressly stated, will be disclaimed by Atlas

PRECAUTIONS

1. Make sure that all electrical wiring has been installed in compliance with the valid regulations.
2. Installation must always be performed by a qualified technician.
3. Installation must be carried out in compliance with the circuit diagrams and connection drawings provided.

NOTE



Some precautions are general and may not apply to your optional equipment.

7.2 OPTIONS

OVERVIEW

1. Safety Valve

- Too high of the pipe network pressure will cause potential safety hazard and damage to the network instrument or actuator and other components. And the Suggestions on the two tower dryer installed safety valves to unload the high pressure of the pipe. If this option is to buy the random, before delivery, the safety valve installed on the device.

2. Vessel bed

- Some high desiccant load working conditions, to prevent when the compressed air go through the cooler with separator, condensed water separation is not completely and the water come into the adsorption tower, drying tower bed needed, drying tower bed nets can better realize the gas water separation, so as to extend the life of the desiccant.

3. Vessel insulation

- To prevent heat loss of the compressor outlet air, the piping in between the compressor hot air outlet and the dryer inlet must be insulated (customer's installation).
To further improve energy efficiency of the dryer, the vessels can be insulated as an option.

4. DDP outlet filter (shipped loose)

- On the outlet side of the dryer, (a) particle filter(s) can be mounted to prevent dust particles, emerging from the desiccant bed, from entering the air net.

The X filter option consists of a DDp outlet filter with alarm contact, as well as all the relevant piping and instructions. An adjustable space of approx. 25 mm (0.98 in) is provided on the support of the filter inlet pipe to account for unevenness in the floor and/or tolerances on the dryer piping. The status of the filter has a read-out on the Elektronikon regulator, which will display an alarm message in case of filter fouling. The filter element(s) of a DDp outlet filter should be replaced one week after initial start-up or replacement of the desiccant.

In case an extra filter is installed (customer's installation), it is important to mount it always downstream of the DDp particle filter.

5. Sonic nozzle

- Excessively high air velocity (e.g. due to low inlet pressure) may cause the desiccant to “float” in the vessels. The desiccant beads scrape against each other, creating additional dust. High-speed flow occurs when a compressor starts with an empty air net or when working at a discharge air pressure that is too low. A high volume of compressed air output can result in a pressure drop, which in turn increases the speed. Therefore, the air net should be filled by slowly opening the valve towards the dryer. Alternatively, a sonic nozzle can be installed downstream of the dryer.

Refer to the instruction delivered with the sonic nozzle to install and adjust it.

6. SST instrument air pipe

- Suitable for heat resistance corrosion environment.

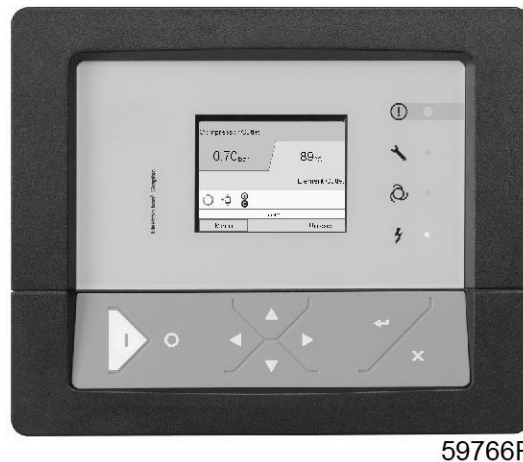
7. PDP option

- Must calibration the dew-point meter every year according to the drying machine cycle to ensure the normal function of dew point reading correct and dew point, About the calibration, please contact with atlas customer service center contact manufacturing

8 PROBLEM SOLVING

8.1 PROBLEM SOLVING

CONTROL PANEL



ATTENTION

	Use only authorized parts. Any damage or malfunction caused by the use of unauthorized parts is not covered by Warranty or Product Liability.
	Before carrying out any maintenance or repairs on the dryer, stop the dryer and switch off the voltage. Open the isolating switch to prevent an accidental start.

When a warning sign pops up, please check the table below to solve the problem.

FAULTS AND REMEDIES

	Warning / Shutdown	Fault	Remedy
1	Poor dew point	High cooling water temperature	Check that condensate separators and drains upstream of the dryer are operative. Check reference
		Ineffective cooler(s) or not enough coolant flow through	Clean cooler(s)
		Working pressure low	Check pressure drop in filter, etc.
		Poor regeneration	See 4
		Dryer fails to shift	See 2 Check compressor outlet

	Warning / Shutdown	Fault	Remedy
		Insufficient water separation in compressor intercooler(s)	Check water drain(s) of compressor intercooler(s)
		Poor desiccant (older than 5 years, contamination in)	Replace desiccant Contact your Atlas Copco
		Low air inlet temperature from compressor	Insulate connection piping if temperature drop is too high
2	Dryer fails to shift	Pilot air missing	Check for pressure in the operating lines. Clean pilot air
		Temperature in one of the vessels above 100 °C (212 °F)	Wait until vessel cools down. (Check and remedy)
		A valve alarm is active	Check alarms and check if the valve is working correctly
		Dewpoint is below defined value	Wait until vessel becomes saturated or disable the PDP
3	Unusually large flow through regeneration valve(s) during	Regeneration inlet valve leaks	Disassemble and check; replace seals
		Incorrect or no nozzle	Check and remedy
4	Regeneration temperature too low	Incorrect working conditions	Check reference conditions and limitations
		Heater fuse or contactor tripped	Check the electrical cabinet
		Insufficient insulation of piping between compressor	Check insulation
5	Too high dryer outlet temperature	Too high adsorption air temperature	Check adsorption air inlet temperature after cooling stage,
		High cooling water temperature	Check that condensate separators and drains upstream of the dryer are operative. Check reference
		Not enough coolant flow through cooler(s)	Check cooling water supply or clean cooler(s)
6	A lot of dust in the air net (The outlet filter element	Desiccant floating due to too low working pressure	Contact your Atlas Copco Customer Centre
		Heavily fluctuating air net	Install sonic nozzle
		Starting up the dryer with empty air net	Install sonic nozzle
		Too much flow through dryer (multiple compressor installation or oversized	Install sonic nozzle
7	Valve fails to change	No pilot air available	Check the air filter of the pilot air.
		Limit switch incorrectly adjusted	Check limit switch

	Warning / Shutdown	Fault	Remedy
8	Dryer makes a lot of noise when dry air cooling	Incorrect or no nozzle installed	Check and remedy
9	EWD on cooler in alarm or not draining	Ball valve to EWD closed	Open valve
		Pressure in dryer lower than 1 bar (14.5 psi)	
		No leds burning	Check fuse and supply Check Printed Circuit Board (PCB) Check if supply on data plate is
		No condensate is drained when pressing test button	Condensate inlet or outlet is blocked EWD requires service
		Condensate is only drained when pressing the test	Leak on cooler Check amount of condensate
		Much condensate coming from compressor	Check if compressor is having many loads/unloads Activate last stage EWD when compressor is in unload. Contact
10	Dryer can not be restarted after shut-down	Dryer in remote control mode	De-activate remote control and select local control Reset and restart the dryer Select
11	Dryer does not start automatically after a power supply interruption	Dryer stopped	Activate the automatic restart function in the Elektronikon regulator
13	Overheating alarm triggered	Heater fuse tripped	Check heater and reset fuse
		Overheating switch tripped	Check and reset

9 TECHNICAL DATA

9.1 REFERENCE CONDITIONS AND LIMITATIONS

REFERENCE CONDITIONS

Condition	Unit	Value
Ambient air temperature	°C	25
Ambient air temperature	°F	27
Ambient relative humidity	%	60
Compressed air effective inlet pressure	bar	7
Compressed air effective inlet pressure	psi	101.5
Compressed air inlet temperature	°C	110
Compressed air inlet temperature	°F	230
Inlet relative humidity of compressed air	%	10%
Cooling water temperature	°C	32.0
Cooling water temperature	°F	80.6
Pressure dewpoint	°C	-40(G-type) / 0 (S-type)
Pressure dewpoint	°F	-40(G-type) / 32 (S-type)

LIMITATIONS

Limitation	Unit	XD 1250	XD 1600	XD 2000	XD 2500	XD 3600	XD 4150	XD 5000
Effective inlet pressure of compressed air								
maximal	bar	10	10	10	10	10	10	10
maximal	psi	145	145	145	145	145	145	145
minimal (2)	bar	See(1)	See(1)	See(1)	See(1)	See	See (1)	See (1)
minimal (2)	psi	See(1)	See(1)	See(1)	See(1)	See	See (1)	See (1)
Ambient air temperature								
maximal	°C	55	55	55	55	55	55	55
maximal	°F	131	131	131	131	131	131	131
minimal	°C	0	0	0	0	0	0	0
minimal	°F	32	32	32	32	32	32	32
Inlet temperature of compressed air								
maximal	°C	130	130	130	130	130	130	130
maximal	°F	266	266	266	266	266	266	266
minimal	°C	90	90	90	90	90	90	90
minimal	°F	194	194	194	194	194	194	194
Max. cooling water temp.								
at inlet	°C	35	35	35	35	35	35	35

at inlet	°F	95	95	95	95	95	95	95
Limitation	Unit	XD 1250	XD 1600	XD 2000	XD 2500	XD 3600	XD 4150	XD 5000
at outlet	°C	45	45	45	45	45	45	45
at outlet	°F	113	113	113	113	113	113	113
Cooling water pressure								
maximal	bar	10	10	10	10	10	10	10
maximal	psi	145	145	145	145	145	145	145
Cooling water flow								
minimal flow	l/s	2x3.84	2x5.13	2x 9.56	2x	2x 9.31	2x 9.31	2x
minimal flow	cfm	2x8.12	2x10.8	2x20.2	2x23.47	2x19.7	2x19.7	2x49.8
Volume flow at dryer inlet (3) (4) (5) (6)								
minimal	Nl/s	375	500	625	750	1060	1250	1500
minimal	cfm	794	1059	1324	1589	2245	2648	3177

Footnotes

- (1) To avoid damage to desiccant, air speed should be limited. Refer to calculation program.
- (2) If effective inlet pressure < 4 bar (58 psi) (and allowed by XD calculation program, available at your Atlas Copco Customer Center), pilot air must be supplied separately because valve control requires > 4 bar(e) (58 psi). Refer to the *Option* section for an available Atlas Copco solution.
- (3) Referred to absolute pressure of 1 bar (14.5 psi) and temperature of 20°C (68°F) and measured according to ISO 7183.
- (4) Flows less than the stated minimum volume flow will reduce general performance due to channelling inside the vessels.
- (5) Flows less than the stated minimum volume flow will reduce regeneration and cooling capacity due to lack of Heat-of-Compression.
- (6) The dryer needs a minimal load time of 4 hours to regenerate and cool. Frequent unloading of the compressor should be prevented.

9.2 AIR DRYER DATA

ZERO PURGE COOLED VERSIONS

Performance data (1)	Unit	XD 1250	XD 1600	XD 2000	XD 2500	XD 3600	XD 4150	XD 5000
Volume flow at dryer Inlet (1) (2) (3)	l/s	1250	1600	2000	2500	3600	4150	5000
Volume flow at dryer Inlet (1) (2) (3)	cfm	2647	3530	4412	5295	7483	8825	10590
Pressure drop over dryer (4)	bar	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Pressure drop over dryer (4)	psi	3.65	3.65	3.65	3.65	3.65	3.65	3.65
- Peak value of max. pressure drop	bar	0.35	0.35	0.35	0.35	0.35	0.35	0.35
- Peak value of max. pressure drop	psi	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Time of half a cycle (G)	min	240	240	240	240	240	240	240

Time of half a cycle (S)	min	240	240	240	240	240	240	240
Performance data (1)	Unit	XD 1250	XD 1600	XD 2000	XD 2500	XD 3600	XD 4150	XD 5000
Power of installed heater (on G-type)	KW	46	62	77	90	120	155	190
Power of installed heater (on G-type)	HP	62.6	84.3	104.7	122.4	163.2	210.8	258.4
Average power consumption (of G-type) (1)	kW	17.3	23.3	28.9	33.8	45	58.1	71.3
Average power consumption (of G-type) (1)	HP	23.5	31.7	39.3	46	61.2	79	97
Average power consumption (of S-type) (1)	kW	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Average power consumption (of S-type) (1)	HP	0.34	0.34	0.34	0.34	0.34	0.34	0.34
Cooling water consumption (5) (6)	lps	3.84	5.13	9.56	11.08	9.31	9.31	23.55
Cooling water consumption (5) (6)	cfm	8.12	10.88	20.24	23.47	19.71	19.71	49.89
Pressure drop over cooling water	bar	1	1	1	1	1	1	1
Pressure drop over cooling water circuit	psi	14.5	14.5	14.5	14.5	14.5	14.5	14.5

Footnotes:

- (1) All performance data stated at reference conditions for standard duty XD variant, with extended cycle time (dew point sensor control). Selection of the dryer at other than reference conditions: refer to the calculation program for XD dryers available at your Atlas Copco Customer Center.
- (2) To avoid damage to the desiccant, air speed should not exceed a given value. Refer to calculation program. It sizes the XD dryer in a way that air speed does not exceed a maximum value.
- (3) Referred to absolute pressure of 1 bar (14.5 psi) and temperature of 20 °C (68 °F) and measured according to ISO 7183.
- (4) Average pressure drop over cycle - pressure drop varies slightly over cycle time.
- (5) Water composition: 100% water, 0% glycol.
- (6) Average water temperature rise of 10 °C (18 °F).

Design Data	Unit	PHCL 750	PHCL 1000	PHCL 1250	PHCL 1500	PHCL 2120	PHCL 2500	PHCL 3000
Desiccant type		AA						
Amount of desiccant per vessel	kg	950	1300	1600	1900	2550	3200	3850
Amount of desiccant per vessel	lbs	2090	2860	3520	4180	5610	7040	8470
Recommended filter size	DDp	1260	1600	2000	2400	3600	4000+	6000+
Mechanical connections								
- Air inlet connection DIN PN16 DN	mm	100	150	150	150	200	200	250
- Dry air outlet connection DIN PN16	mm	100	150	150	150	200	200	250
- Cooling water inlet and outlet connection DIN PN16 DN	mm	50	50	50	50	80	80	100
Dimensions								
- Length	mm	3500	3780	3780	4123	4750	5150	6123
- Width	mm	2548	2758	2848	2848	3178	3450	3912
- Height	mm	2850	3018	3131	3172	3537	3605	4062

10 DOCUMENTATION

DECLARATION OF CONFORMITY

Typical example of a Declaration of Conformity document



EC DECLARATION OF CONFORMITY

- 1 We, Atlas Copco Airpower n.v., declare under our sole responsibility, that the product
- 2 Machine name
- 3 Machine type
- 4 Serial number
- 5
- 6 Which falls under the provisions of article 12.2 of the EC Directive 2006/42/EC on the approximation of the laws of the Member States relating to machinery, is in conformity with the relevant Essential Health and Safety Requirements of this directive.

The machinery complies also with the requirements of the following directives and their amendments as indicated.

7	Directive on the approximation of laws of the Member States relating to		Harmonized and/or Technical Standards used	Att'ment
a.	Pressure equipment	97/23/EC		
b.	Machinery safety	2006/42/EC	EN ISO 12100 – 1 EN ISO 12100 – 2 EN 1012 – 1	
c.	Simple pressure vessel	2009/105/EC		
d.	Electromagnetic compatibility	2004/108/EC	EN 61000-6-2 EN 61000-6-4	
e.	Low voltage equipment	2006/95/EC	EN 60034 EN 60204-1 EN 60439	
f.	Outdoor noise emission	2000/14/EC		
g.	Equipment and protective systems in potentially explosive atmospheres	94/9/EC		
h.	Medical devices	93/42/EEC	EN ISO 13485 EN ISO 14971 EN ISO 7396	
i.				

- 8.a The harmonized and the technical standards used are identified in the attachments hereafter
- 8.b Atlas Copco Airpower n.v. is authorized to compile the technical file.
- 9
- 10
- 11
- 12
- 13
- 14
- 15
- 16
- Conformity of the specification to the directives
- Conformity of the product to the specification and by implication to the directives
- Issued by Engineering Manufacturing
- Name
- Signature
- Date

Form 5009 xxxx-xx
ed. xx, xxxx -xx-xx

Atlas Copco Airpower n.v.

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representative

What sets Atlas Copco apart as a company is our conviction that we can only excel in what we do if we provide the best possible know-how and technology to really help our customers produce, grow and succeed.

There is a unique way of achieving that - we simply call it the Atlas Copco way. It builds on **interaction**, on long-term relationships and involvement in the customers' process, needs and objectives. It means having the flexibility to adapt to the diverse demands of the people we cater for.

It's the **commitment** to our customers' business that drives our effort towards increasing their productivity through better solutions. It starts with fully supporting existing products and continuously doing things better, but it goes much further, creating advances in technology through **innovation**. Not for the sake of technology, but for the sake of our customer's bottom line and peace-of-mind.

That is how Atlas Copco will strive to remain the first choice, to succeed in attracting new business and to maintain our position as the industry leader.